Hyperelliptic sigma solutions of polynomial Hamiltonian systems

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Working draft: October 2026

Abstract

We study polynomial Hamiltonian systems whose Hamiltonians consist of two weighted products of canonical modes minus a polynomial in their powers. This class extends the structure of familiar genus-one coupled-mode systems from nonlinear optics. We construct local analytic solutions in terms of sigma functions and Abelian integrals on smooth odd-degree hyperelliptic curves, subject to stated degree and regularity conditions. The paired powers and complete weighted mode products are related to Ônishi's determinant identities, extending the Frobenius–Stickelberger structure of the elliptic case. Explicit genus-two and genus-three examples include bounded oscillatory scalar motion. We evaluate the solutions using Generapy, a purpose-built Python package for algebraic curves and Abelian functions, and validate them against independent fourth-order Runge–Kutta integration.

1 Introduction

Many algebraic coupled-mode systems in nonlinear optics admit genus-one elliptic solutions. Examples include four-wave mixing, polarization dynamics and coherent couplers in cubic media, and two-mode and three-mode mixing in quadratic media. These systems are characterized by a conserved polynomial Hamiltonian and Manley–Rowe-like intermodal power laws. Together with their elliptic reductions, these conservation laws reveal a common mathematical structure [1, 2, 3, 4, 5, 6, 7, 8, 9].

The same viewpoint includes the polynomial differential systems defining Jacobi, lemniscatic and Dixon elliptic functions [10]. The coupled-mode systems also admit projections onto the unforced, undamped Duffing oscillator, the Euler top and the Lagrange top. These connections link optical models to the classical and geometric mechanics of integrable systems; background accounts are given in [11, 12, 13, 14, 15].

Optical solutions commonly express modal powers through Jacobi or Weierstrass elliptic functions and recover phases through elliptic integrals. In [7, 8, 9], the full complex amplitudes were instead expressed as shifted Weierstrass sigma ratios with exponential factors. Their conservation laws have a common determinant origin: the 2×2 Frobenius–Stickelberger identity gives paired-mode powers, while higher-order determinants give the complete products entering the Hamiltonian [17, 8, 9].

The mode count and interaction multiplicities are integer parameters; in optics, the interaction degree is related to the material nonlinearity. Increasing these parameters can produce scalar curves of genus greater than one. What then replaces the elliptic sigma solutions and their determinant conservation laws?

Ônishi's identities [16] generalize the Frobenius–Stickelberger determinant to hyperelliptic curves of arbitrary genus. Their 2×2 specialization expresses a difference of curve coordinates as a product of oppositely shifted sigma factors, matching the affine dependence of paired-mode powers on a common scalar coordinate. Eliminating that coordinate between two such powers gives a conserved weighted difference, the higher-genus counterpart of a Manley–Rowe-like intermodal power law. The multipoint identity suggests the corresponding complete mode

products. This raises the question of whether these sigma factors and determinants provide the mode solutions and conservation laws for higher-genus canonical systems.

The theory of hyperelliptic and, more broadly, Abelian functions has its roots in the nineteenth-century work of Abel, Jacobi, Göpel, Rosenhain, Weierstrass, Riemann, Klein and Poincaré, among others. Baker's books at the turn of the twentieth century brought together and substantially enhanced much of this theory [18, 19]. A renewed connection with nonlinear integrable equations emerged in the 1970s through finite-gap integration and algebro-geometric solutions, in work by Novikov, Dubrovin, Its, Matveev, Krichever and others [20, 21]. For historical background and introductions to sigma functions, we refer the reader to Baker's books, the review by Buchstaber, Enolski and Leykin [22], Matsutani's monograph [23] and the references therein.

Hyperelliptic Kleinian functions provide an algebraic framework for Jacobi inversion and integrable equations, as developed by Buchstaber, Enolskii and Leykin [24]. For the scalar motion considered here, the relevant inversion takes place on the one-point Abel image rather than at a general point of the Jacobian. Matsutani and Previato [25, 26] study inversion and sigma-derivative identities on these lower-dimensional strata for cyclic curves, including the odd-degree hyperelliptic family. Enolski and collaborators [27, 28] develop sigma-based inversion methods and apply them to relativistic particle motion, including black-hole space-times. Their third-kind integral formulas, together with the hyperelliptic prime-form representation of Gibbons, Matsutani and Ônishi [29], provide tools relevant to individual-mode reconstruction.

The complete mode products also connect with algebraic interpolation and addition laws. Bernatska [30] constructs curve functions with prescribed zeros through normalized interpolation determinants, while Bernatska and Kopeliovich [32] use hyperelliptic interpolation to describe divisor addition and reduction. These constructions clarify the relation between the determinants, their complementary zeros and the polynomial interpolation used herein. Beyond arguments restricted to individual curve points, Eilbeck, Enolskii and Previato [33] derive a genus-two Frobenius–Stickelberger addition formula for three general Jacobian arguments.

1.1 Our contribution

We consider N independent canonical pairs of complex-valued functions $u_j(z), v_j(z)$ of an evolution variable z, with Hamiltonian

$$H(u,v) = \prod_{j=1}^{N} u_j^{\beta_j} + \prod_{j=1}^{N} v_j^{\beta_j} - Q(I_1, \ldots, I_N),$$

where $I_j(z) = u_j(z)v_j(z)$ are generally complex quantities, the weights β_j are positive integers, and Q is a polynomial in the I_j. We call these paired products modal powers, without assuming that they are real or positive. Manley–Rowe-like conservation laws relate their evolution through conserved weighted differences, stated in (3). Together with the Hamiltonian, these laws reduce the power dynamics to a single scalar equation on a hyperelliptic curve. Recovering the individual modes requires further integration, since their paired products alone do not determine them.

This reduction connects the systems to Martin Kummer's resonant Hamiltonians [34, 11] and the Kummer shape algebras of Odzijewicz and Wawreniuk [35].[1] The conserved weighted differences arise from symmetries of the canonical modes, while the products entering the Hamiltonian describe their collective interaction.

Our main contribution is a construction of local solutions for the individual complex modes using hyperelliptic sigma functions and Abelian integrals. We connect both the paired

[1] Here Kummer refers to Martin Kummer, distinct from Ernst Eduard Kummer of algebraic number theory and algebraic geometry.

modal powers and the complete weighted products in the Hamiltonian to Ônishi's determinant identities. In particular, we identify how the individual mode factors combine into these determinants and why their exponential corrections cancel in the complete products. Explicit genus-two and genus-three examples illustrate the construction, including bounded oscillatory scalar motion in genus three. We evaluate the sigma solutions numerically and compare them with independent integration of the full canonical equations.

We evaluate the analytic solutions using Generapy [36], a purpose-built Python library for algebraic curves and Abelian functions, which supplies periods, Abelian integrals and sigma functions. Generapy is built on mpmath [37] for arbitrary-precision numerical arithmetic. During its development, we tested Generapy by reproducing several examples from the literature, including the period and Kleinian-function computations for Bernatska's genus-three trigonal curve [31]; the reproduced examples are described in the Generapy documentation [36].

Sections 2–4 develop scalar reduction, inversion and mode reconstruction; Section 5 establishes the determinant connection. Sections 6 and 7 present the examples and comparisons with independent fourth-order Runge–Kutta integration. The solutions are local and apply to smooth odd-degree hyperelliptic curves, subject to the degree and regularity conditions stated in the main results.

2 Scalar reduction and separated mode equations

We start from the polynomial Hamiltonian and derive the common scalar motion of the modal powers. The same calculation expresses each mode equation in terms of that scalar trajectory and separates its logarithmic derivative into a fixed-point contribution and a polynomial correction. These are the two terms integrated in the following sections.

2.1 Hamiltonian equations and conserved powers

Let $u_j(z), v_j(z) \in \mathbb{C}$, $j = 1, \dots, N$, be independent complex canonical modes, with primes denoting derivatives with respect to z. Define

$$I_j = u_j v_j, \qquad U = \prod_{j=1}^{N} u_j^{\beta_j}, \qquad V = \prod_{j=1}^{N} v_j^{\beta_j},$$

where the weights β_j are positive integers. We consider

$$H(u, v) = U + V - Q(I_1, \dots, I_N), \tag{1}$$

with Q a polynomial in the powers. Here "power" denotes $I_j = u_j v_j$ and need not be real or positive. Write $Q_j = \partial Q / \partial I_j$, treating the powers as independent arguments of Q before evaluating them along a trajectory. Hamilton's equations are

$$\begin{aligned} u_j' &= \frac{\partial H}{\partial v_j} = \beta_j v_j^{\beta_j - 1} \prod_{\substack{1 \le k \le N \\ k \ne j}} v_k^{\beta_k} - Q_j(I) u_j, \\ v_j' &= -\frac{\partial H}{\partial u_j} = -\beta_j u_j^{\beta_j - 1} \prod_{\substack{1 \le k \le N \\ k \ne j}} u_k^{\beta_k} + Q_j(I) v_j. \end{aligned} \tag{2}$$

The Hamiltonian is conserved because it has no explicit dependence on z. Multiplying the first equation by v_j and the second by u_j cancels the terms containing Q_j and gives

$$I_j' = \beta_j (V - U).$$

Thus every normalized power I_j/β_j has the same derivative. In particular, the $N-1$ independent weighted differences

$$C_j = \beta_N I_j - \beta_j I_N, \qquad j = 1, \dots, N-1, \tag{3}$$

are conserved. These are the Manley–Rowe-like intermodal power laws of the canonical system.

2.2 A common scalar coordinate

The conserved differences allow all powers to be written using a single scalar coordinate. Fix the additive freedom in the scalar coordinate by defining

$$\rho = -\frac{\sum_{j=1}^{N} I_j}{\sum_{j=1}^{N} \beta_j}, \qquad \gamma_j = I_j + \beta_j \rho.$$

The power equations imply

$$\rho' = U - V, \qquad \gamma_j' = 0, \qquad \sum_{j=1}^{N} \gamma_j = 0.$$

Consequently

$$I_j = \gamma_j - \beta_j \rho, \tag{4}$$

and the conserved differences take the form

$$\beta_k I_j - \beta_j I_k = \beta_k \gamma_j - \beta_j \gamma_k.$$

The constants γ_j are therefore fixed by the initial powers, while ρ describes their common evolution. Equivalently, $-\rho$ is the weighted mean of the normalized powers I_j/β_j, with weights β_j.

On this power line, define the polynomial restrictions

$$\begin{aligned} q(\rho) &= Q(\gamma_1 - \beta_1 \rho, \dots, \gamma_N - \beta_N \rho), \\ r(\rho) &= \prod_{j=1}^{N} (\gamma_j - \beta_j \rho)^{\beta_j} = UV. \end{aligned} \tag{5}$$

The constant term of Q sets the zero of energy without changing the equations of motion. We choose it so that the trajectory under consideration lies on $H = 0$, giving $U + V = q(\rho)$. Combining the sum and difference of U, V gives

$$U = \frac{q(\rho) + \rho'}{2}, \qquad V = \frac{q(\rho) - \rho'}{2}, \tag{6}$$

and therefore

$$(\rho')^2 = (U - V)^2 = (U + V)^2 - 4UV = q(\rho)^2 - 4r(\rho) = F(\rho). \tag{7}$$

This is the scalar curve governing the modal powers. Here and below, q_ρ and F_ρ denote polynomial derivatives with respect to ρ. Along a nonconstant trajectory the scalar equation also gives

$$\rho'' = \frac{1}{2} F_\rho(\rho).$$

It also follows directly by differentiating $\rho' = U - V$ with Hamilton's equations, so it remains valid at turning points, where $\rho' = 0$. It determines continuation there without choosing a square-root sign anew.

2.3 Separated equations for the individual modes

Knowing the powers does not yet determine the individual modes. On a chart where $I_j \neq 0$, divide the canonical equations by u_j, v_j and substitute the product formulas above:

$$\begin{aligned} \frac{u_j'}{u_j} &= \frac{\beta_j V}{I_j} - Q_j(I) = \frac{q(\rho) - \rho'}{2[\gamma_j/\beta_j - \rho]} - Q_j(I(\rho)), \\ \frac{v_j'}{v_j} &= -\frac{\beta_j U}{I_j} + Q_j(I) = \frac{-q(\rho) - \rho'}{2[\gamma_j/\beta_j - \rho]} + Q_j(I(\rho)). \end{aligned} \tag{8}$$

Each right-hand side depends only on the scalar trajectory ρ, ρ'. Once that trajectory is known, the modes follow by separate logarithmic integrations, with initial amplitudes inherited from the original canonical system.

The constants γ_j/β_j and $q(\gamma_j/\beta_j)$ can be expressed as coordinates of fixed points on the scalar curve. Introduce μ_j by

$$\rho(\mu_j) = \frac{\gamma_j}{\beta_j}, \qquad \rho'(\mu_j) = q(\rho(\mu_j)). \tag{9}$$

Here y denotes the independent ordinate on the curve $y^2 = F(\rho)$; along the moving trajectory it is $\rho'(z)$. The notation μ_j labels the fixed point with the displayed coordinates, which lies on the curve because $r(\rho(\mu_j)) = 0$. Adding and subtracting the fixed polynomial value gives

$$\begin{aligned} \frac{u_j'}{u_j} &= \frac{\rho'(\mu_j) - \rho'}{2[\rho(\mu_j) - \rho]} + B_j(\rho), \\ \frac{v_j'}{v_j} &= \frac{-\rho'(\mu_j) - \rho'}{2[\rho(\mu_j) - \rho]} - B_j(\rho), \end{aligned} \tag{10}$$

where

$$B_j(\rho) = \frac{q(\rho) - q(\rho(\mu_j))}{2[\rho(\mu_j) - \rho]} - Q_j(I(\rho)). \tag{11}$$

The divided difference is polynomial because its numerator vanishes at $\rho = \rho(\mu_j)$; the restricted partial derivative is polynomial as well. Thus B_j is polynomial, determined by the original Hamiltonian. In particular, differentiating Q before restricting it to the power line retains information needed for individual modes that is not contained in q alone.

2.4 Hyperelliptic scope of this paper

The Hamiltonian reduction above is algebraic and does not require a smooth hyperelliptic curve or a degree bound on B_j. To express its separated integrals in a manageable hyperelliptic form, we initially focus on the following setting:

- F is squarefree of odd degree $2g + 1$, with $g \geq 2$ and leading coefficient 4.
- The fixed coordinates $\rho(\mu_j)$ are distinct and $q(\rho(\mu_j)) \neq 0$.
- Every polynomial correction satisfies $\deg B_j \leq g - 1$.

The first condition gives a smooth compactified hyperelliptic curve of genus g with one point at infinity. The leading coefficient can always be normalized to 4 by rescaling z and the modes u_j, v_j; we use this normalization throughout. The second keeps the fixed mode points distinct and away from branch points. The third makes holomorphic Abel differentials sufficient for the polynomial corrections: writing

$$B_j(\rho) = \sum_{k=1}^{g} c_{jk}\rho^{k-1}, \qquad \deg B_j \le g-1, \tag{12}$$

we have $dz = d\rho/y$ along the trajectory, so each term integrates through $\rho^{k-1}d\rho/y$. The remaining rational point terms will be integrated using shifted sigma ratios with exponential corrections.

The degree condition holds automatically if $\deg Q \le g$. For higher-degree Q, form B_j and check that every coefficient of ρ^g and higher vanishes. Both worked examples satisfy this test through cancellation of leading terms.

These restrictions define the initial scope of the analytic construction, rather than limits of the Hamiltonian reduction. Higher-degree corrections lead to meromorphic Abelian differentials; even-degree models, singular curves and coincident or branch fixed points require corresponding changes to the curve representation or limiting arguments. We do not develop those extensions here.

The reduction has now supplied the scalar curve, its fixed mode points and the two terms in every mode integral. The following sections first recover the scalar trajectory and then integrate these separated equations. The logarithmic chart excludes modal zeros, but the original polynomial canonical equations remain the equations for continuation of a finite solution through such points.

3 Scalar inversion and its hyperelliptic sigma representation

Section 2 reduced the modal powers to $(\rho')^2 = F(\rho)$. Its local analytic solution is the inverse of a hyperelliptic Abelian integral. We first describe that inverse and its Abel coordinates, then give a sigma-derivative representation of the scalar coordinate. The same Abel coordinates and sigma conventions will be used to reconstruct the individual modes.

3.1 The scalar solution as an inverse Abelian integral

Let the smooth compactification of

$$y^2 = F(\rho) = 4\rho^{2g+1} + \cdots \tag{13}$$

have genus $g \ge 2$, with its unique point at infinity. Choose an initial finite curve point $(\rho(z_0), \rho'(z_0))$. The scalar solution is determined locally by

$$z - z_0 = \int_{(\rho(z_0),\rho'(z_0))}^{(\rho(z),\rho'(z))} \frac{d\rho}{y}. \tag{14}$$

The integral is taken along the continued curve path, so its sheet is fixed by the initial point. The differential $d\rho/y$ is nonzero at every finite curve point, including branch points when written in a regular local parameter. Its integral therefore has a local analytic inverse. At a finite turning point, where $\rho' = 0$, the equation $\rho'' = F_\rho/2$ continues the signed trajectory through the point. At infinity the differential has a zero of order $2g-2$, so no global single-valued inverse in z is asserted.

This inverse integral defines the scalar analytic solution. The sigma formulas below represent its coordinate on the Abel image; evaluating them at prescribed z still requires determining the corresponding point of that image.

3.2 Abel coordinates and propagation

The holomorphic differentials and the Abel lift of the moving point are

$$\omega_k = \frac{\rho^{k-1}\,d\rho}{y}, \qquad k = 1, \dots, g, \tag{15}$$

$$w_k(z) = \int_{\infty}^{(\rho(z),\rho'(z))} \omega_k. \tag{16}$$

Choose paths from infinity to the initial moving point and the fixed mode points, and continue the moving path with the trajectory. The Abel vectors are lifts, rather than vectors reduced modulo periods; subsequent sigma expressions and second-kind integrals use compatible path choices. Along the trajectory,

$$w_1(z) = w_1(z_0) + z - z_0, \qquad w_k'(z) = \rho(z)^{k-1}. \tag{17}$$

The initial offset is retained because the Abel map is based at infinity, whereas z_0 is an arbitrary propagation origin. Only w_1 is linear in propagation. For genus one this sole coordinate is enough to evaluate the familiar elliptic solution directly. For $g \geq 2$, the additional coordinates must lie with w_1 on the one-dimensional Abel image W_1 of the curve. They cannot be replaced by independently chosen linear functions of z.

3.3 Sigma and second-kind normalization

Write $F(\rho) = \sum_{m=0}^{2g+1} \lambda_m \rho^m$, where $\lambda_{2g+1} = 4$, and put $\lambda_m = 0$ outside this range. We use the algebraic second-kind basis [28, Eq. II.2]

$$dr_k = \frac{1}{4y} \sum_{m=k}^{2g+1-k} (m+1-k)\lambda_{m+1+k}\rho^m\,d\rho, \qquad k = 1, \dots, g. \tag{18}$$

Together with (15), this fixes the Kleinian sigma exponential convention. More explicitly, let $2\omega, 2\omega'$ be the first-kind period matrices and $2\eta = -\oint_a dr$ the second-kind a-period matrix in the same homology basis. Then

$$\sigma(w) = C \exp\!\left(\tfrac{1}{2} w^T \kappa w\right) \theta[\delta]\big((2\omega)^{-1}w; \tau\big), \quad \kappa = \eta\omega^{-1}, \quad \tau = \omega^{-1}\omega', \tag{19}$$

where δ is the characteristic of the Riemann constant based at infinity. The constant C fixes the leading Schur–Weierstrass polynomial. We use Ônishi's sigma normalization [16], as implemented in generapy:

$$\sigma(w) = w_1 - \tfrac{1}{3}w_2^3 + \text{higher-weight terms}, \qquad g = 2, \tag{20}$$

$$\sigma(w) = w_1 w_3 - w_2^2 - \tfrac{1}{3}w_2 w_3^3 + \tfrac{1}{45}w_3^6 + \text{higher-weight terms}, \qquad g = 3. \tag{21}$$

The polynomial for genus three also agrees with [27, Eq. 3.39]. References to multipoint determinant identities are understood with Ônishi's published errata. The two-point addition identity used below retains the sign stated in [16, Lemma 7.1].

Let W_n denote the Abel image of effective degree-n divisors, based at infinity. For $g \geq 2$, $W_1 \subset W_{g-1}$, and W_{g-1} is the sigma zero divisor. Consequently $\sigma(w(z)) = 0$. In genus three the one-point image also satisfies $\sigma_3 = \sigma_{33} = 0$ [28, Section VI].

Sigma subscripts denote partial derivatives in the Abel coordinates. The derivatives used for the mode kernel are [16]

$$\sigma_\sharp = \left(\prod_{\substack{2\leq k\leq g \\ k \text{ even}}} \partial_{w_k}\right)\sigma, \qquad \sigma_\flat = \left(\prod_{\substack{3\leq k\leq g \\ k \text{ odd}}} \partial_{w_k}\right)\sigma. \tag{22}$$

An empty product takes no derivative: $(\sigma_\sharp, \sigma_\flat) = (\sigma_2, \sigma)$ for genus two and (σ_2, σ_3) for genus three. These kernel factors are distinct from the derivative ratios used to recover ρ.

3.4 From Abel inversion to a sigma coordinate ratio for $\rho(z)$

The inverse integral (14) determines the moving curve point. We now express its coordinate through sigma functions evaluated at its Abel vector. This is the one-point restriction of Jacobi inversion: the arguments lie on W_1, rather than being arbitrary points of the Jacobian.

Ordinary Jacobi inversion starts with the summed Abel vector of g curve points and recovers their coordinates. Our trajectory contains only one moving point, so we use a limiting restriction of that problem. Sending $g-1$ divisor points to infinity leaves a single finite point. Ordinary sigma quotients can then have vanishing numerators and denominators, so one must expand in local coordinates at infinity and retain the first nonzero derivatives. Hyperelliptic addition and determinant identities control these limits. The resulting derivative indices are determined by the gap sequence at infinity, not by a guess made separately for each genus.

For the differential ordering (15), the general result is particularly simple. Define

$$\mathcal{D}_g = \prod_{\substack{4\le k\le g\\ k \text{ even}}} \partial_{w_k}. \tag{23}$$

The product is empty in genera two and three. Since $\sigma_\sharp = \partial_{w_2}\mathcal{D}_g\sigma$, the scalar inversion theorem on the one-point stratum gives

Proposition 3.1 (One-point sigma coordinate recovery)**.** *For $g \ge 2$ and $w(z)$ on the one-point Abel image, on a chart where the denominator is nonzero,*

$$\rho(z) = -\frac{\partial_{w_1}\mathcal{D}_g\sigma(w(z))}{\partial_{w_2}\mathcal{D}_g\sigma(w(z))}. \tag{24}$$

Proof. In the notation of [29, Definition 5.1 and Theorem 5.5], the one-point derivative index is $\natural_1 = \{2,4,6,\dots\} \cap \{1,\dots,g\}$. Replacing its index 2 by 1 gives $\natural_1^{(1)}$. The $k=1$ stratum inversion theorem states that the ratio of these derivatives is minus the coordinate of the single finite point. These two derivatives are precisely $\partial_{w_1}\mathcal{D}_g\sigma$ and $\partial_{w_2}\mathcal{D}_g\sigma$. An overall multiplicative normalization of sigma cancels from this ratio. Replacing the monic ordinate by $y/2$ gives the same Abel differential basis after the translation to our curve convention. □

The formula represents the scalar coordinate analytically on its constrained Abel image. Its origin can also be seen directly in the two-point addition identity [16, Lemma 7.1]. If $w = w(z)$ and $w_\infty(t)$ is the Abel vector of a point with $\rho = t^{-2}$ tending to infinity, then

$$\begin{aligned}
(-1)^{g+1}\frac{\sigma_\flat(w+w_\infty(t))\sigma_\flat(w-w_\infty(t))}{\sigma_\sharp(w)^2\sigma_\sharp(w_\infty(t))^2} &= t^{-2} - \rho(z),\\
w_{\infty,k}(t) &= -\frac{t^{2g-2k+1}}{2g-2k+1} + O(t^{2g-2k+3}).
\end{aligned} \tag{25}$$

Subtracting the known t^{-2} pole and extracting the finite part recovers $-\rho(z)$. Expanding the sigma functions makes this a finite-derivative procedure; the stratified inversion theorem identifies the shorter ratio (24). Sending only one auxiliary point to infinity in the two-point identity gives this finite-part construction, while the full Jacobi-divisor restriction sends $g-1$ points to infinity.

3.5 Explicit coordinate ratios in genera two, three and four

The general ratio becomes $-\sigma_1/\sigma_2$ for both genera two and three, and $-\sigma_{14}/\sigma_{24}$ for genus four. The following equivalent low-genus representations are useful in the conventions of Enolski and collaborators [27, 28]:

$$\rho(z) = -\frac{\sigma_1(w(z))}{\sigma_2(w(z))}, \qquad g = 2, \tag{26}$$

$$\rho(z) = -\frac{\sigma_{13}(w(z))}{\sigma_{23}(w(z))}, \qquad g = 3, \tag{27}$$

$$\rho(z) = -\frac{\sigma_{144}(w(z))}{\sigma_{244}(w(z))}, \qquad g = 4. \tag{28}$$

All are evaluated on the one-point image, with nonzero displayed denominators. In genus two it is characterized locally by $\sigma = 0$; in genus three the relevant branch satisfies $\sigma = \sigma_3 = 0$; in genus four it satisfies $\sigma = \sigma_4 = \sigma_{44} = 0$. The genus-four formula is given explicitly in [28, Eq. VII.18]. These higher-derivative formulas are alternative representations on the same stratum, rather than the definition of the arbitrary-genus ratio.

For genus two, differentiating $\sigma(w(z)) = 0$ gives $\sigma_1 + \rho\sigma_2 = 0$. In genus three $\sigma_3 = 0$ removes the last term when differentiating $\sigma = 0$, giving the same first-derivative ratio; differentiating $\sigma_3 = 0$ and using $\sigma_{33} = 0$ gives the displayed alternative (27). Such identities explain why different derivative orders can represent the same coordinate.

3.6 Evaluation at prescribed propagation

The sigma ratio represents ρ on the one-point Abel image, where only w_1 is linear in propagation. Evaluation at prescribed z therefore also requires determining the constrained Abel vector. In our computations, numerical inversion of (14) supplies $\rho(z)$, after which Abelian integrals and sigma evaluations reconstruct the individual modes. The scalar sigma ratio serves as a consistency check.

3.7 Fixed mode points and the mode differential

The fixed mode points were defined in (9). The numerical trajectory need not visit these fixed points. The fixed lift $w(\mu_j)$ is calculated directly from the curve point, using paths compatible with (16).

The mode differential is

$$\omega_j^{\mathrm{mode}} = \frac{\rho'(\mu_j) - y}{2[\rho(\mu_j) - \rho]} \frac{d\rho}{y}. \tag{29}$$

It has residue $+1$ at $(\rho(\mu_j), -\rho'(\mu_j))$ and residue -1 at infinity, with a removable singularity at the fixed point of (9). Along the moving point it is exactly the rational term of u_j'/u_j in (10) times dz. The next section integrates this differential through the sharp/flat sigma factors and combines it with the polynomial correction B_j.

4 Individual mode solutions

Once the scalar trajectory is known, the remaining task is to recover the individual amplitudes rather than just their products. We integrate the fixed-point contribution by sigma functions, determine its exponential correction, and then combine it with the polynomial correction to obtain the full canonical solution.

4.1 The fixed-point second-kind coefficients

In arbitrary genus, define

$$h_{jk} = -\operatorname{Reg} \int_{\infty}^{(\rho(\mu_j),\rho'(\mu_j))} dr_k. \tag{30}$$

Here Reg subtracts the divergent powers of the local coordinate $t = \rho^{-1/2}$ at infinity and chooses zero constant term in the primitive there. The sheet and the lifted path are the same as for $w(\mu_j)$. These choices fix the integration constants and signs; h_{jk} depend on the fixed point and not on propagation.

In genera two and three, put $\zeta_{\sharp,k} = \partial_{w_k} \log \sigma_\sharp$. The coefficients have the short forms

$$\begin{aligned} h_{j2} &= \tfrac{1}{2}\zeta_{\sharp,2}(w(\mu_j)), \\ h_{j1} &= \zeta_{\sharp,1}(w(\mu_j)) - \frac{\sigma_1(w(\mu_j))}{\sigma_2(w(\mu_j))} h_{j2}, \\ h_{j3} &= \zeta_{\sharp,3}(w(\mu_j)) \qquad (g = 3). \end{aligned} \tag{31}$$

In genus two $\sigma_1/\sigma_2 = -\rho(\mu_j)$, so $h_{j1} = \zeta_{\sharp,1} + \rho(\mu_j)h_{j2}$. The genus-three expressions are the negatives of the second-kind primitives in [28, Proposition III.2, Eq. III.10]; the finite-base constants there are replaced by the zero finite part at infinity in (30). The same infinity prescription gives the first two expressions in genus two. These short expressions are convenient low-genus evaluations; the regularized-integral definition applies in every genus, as does the finite sigma-based coefficient extraction described after Lemma 4.1.

4.2 The sigma logarithmic differential

Lemma 4.1. *With the first- and second-kind bases and sigma normalization above, define on compatible regular lifts the uncorrected point factors (denoted by a bar)*

$$\bar{u}_j(z) = \frac{\sigma_\flat(w(z) + w(\mu_j))}{\sigma_\sharp(w(z))\sigma_\sharp(w(\mu_j))}, \qquad \bar{v}_j(z) = \frac{\sigma_\flat(w(z) - w(\mu_j))}{\sigma_\sharp(w(z))\sigma_\sharp(w(\mu_j))}.$$

Then

$$\frac{d}{dz} \log \bar{u}_j(z) = \frac{\rho'(\mu_j) - \rho'(z)}{2[\rho(\mu_j) - \rho(z)]} + \sum_{k=1}^{g} h_{jk} \rho(z)^{k-1}. \tag{32}$$

For $\bar{v}_j$ replace $\rho'(\mu_j)$ and h_{jk} by their negatives. Moreover,

$$\bar{u}_j(z)\bar{v}_j(z) = (-1)^{g+1}[\rho(\mu_j) - \rho(z)]. \tag{33}$$

Proof. We describe the normalization translation behind the third-kind formula. On the lifted curve, the sharp/flat kernel is the ratio of the algebraic prime form between the moving point and the sheet involution of the fixed point to its infinity factors. This is the hyperelliptic prime-form representation of [29, Theorem 7.17]. To translate its monic curve convention to ours, use the ordinate $y/2$; the holomorphic basis $d\rho/(2(y/2))$ is then (15). The second-kind basis is fixed by (18), rather than by an independent holomorphic shift.

The logarithmic kernel differential and (29) have the same simple poles and residues; their difference is a holomorphic differential in the moving point. The sigma quasi-period multipliers have logarithmic derivatives given by periods of dr_k, so compatible continuation preserves this difference in the moving lift. Write the difference as $\sum_{k=1}^{g} h_{jk}\omega_k$.

To identify its coefficients, vary the fixed curve point, holding the moving point fixed. The mixed logarithmic derivative of the algebraic prime form is the algebraic fundamental bidifferential. In [28, Eq. II.11], that bidifferential is the derivative of the rational Cauchy

differential plus $dr^T\omega$. Evaluating the fixed argument on its opposite sheet changes dr to $-dr$. Subtracting the derivative of the rational mode differential therefore gives

$$d_{\mu_j}\big(d_z \log \bar{u}_j(z) - \omega_j^{\mathrm{mode}}\big) = -\sum_{k=1}^{g} dr_k(\mu_j)\,\omega_k(z), \qquad d_{\mu_j} h_{jk} = -dr_k(\mu_j).$$

Here d_{μ_j} denotes variation along the fixed-point copy of the curve, not a propagation derivative. Integrating in the fixed point leaves an additive constant for each coefficient. The infinity prescription subtracts the divergent local powers and sets the remaining constant term to zero; compatible sheet reversal gives the corresponding negative primitive. These choices yield (30). Substituting $\omega_k = \rho^{k-1}dz$ proves (32). Applying the sheet involution gives the formula for $\bar{v}_j$.

The product identity is Ônishi's two-point addition formula [16, Lemma 7.1]. The factor $(-1)^{g+1}$ in (33) is the prefactor of that identity in the stated normalization. This is the infinity-based form of the third-kind integration formula; Eq. III.17 of [28] instead uses ordinary sigma with finite-base shifts. □

Alternatively, in any genus, the coefficients h_{jk} can be recovered directly from the sigma kernel. Subtracting the rational term in (32) from its left-hand side leaves a polynomial in ρ of degree at most $g-1$. Its coefficients are determined by evaluations at g regular curve points with distinct ρ-coordinates, or by its first g Taylor coefficients in ρ at one regular nonbranch point. The points must avoid poles and zero sigma denominators, and all evaluations use compatible Abel lifts. This is coefficient extraction from an exact polynomial identity, rather than a fit to the mode trajectory; the low-genus formulas above give shorter evaluations of the same constants.

4.3 Point factors and polynomial corrections

Put $\Delta w_k(z) = w_k(z) - w_k(z_0)$. The corrected point factors are

$$\begin{aligned}\widehat{u}_j(z) &= \sqrt{\beta_j}\,\bar{u}_j(z)\exp\left[-\sum_{k=1}^{g} h_{jk}\Delta w_k(z)\right],\\ \widehat{v}_j(z) &= (-1)^{g+1}\sqrt{\beta_j}\,\bar{v}_j(z)\exp\left[\sum_{k=1}^{g} h_{jk}\Delta w_k(z)\right].\end{aligned} \tag{34}$$

Lemma 4.1 gives

$$\begin{aligned}\frac{\widehat{u}_j'}{\widehat{u}_j} &= \frac{\rho'(\mu_j)-\rho'(z)}{2[\rho(\mu_j)-\rho(z)]},\\ \frac{\widehat{v}_j'}{\widehat{v}_j} &= \frac{-\rho'(\mu_j)-\rho'(z)}{2[\rho(\mu_j)-\rho(z)]},\end{aligned} \tag{35}$$

and

$$\widehat{u}_j(z)\widehat{v}_j(z) = \beta_j[\rho(\mu_j)-\rho(z)] = I_j(z). \tag{36}$$

Consequently, with continued logarithms,

$$\int_{(\rho(z_0),\rho'(z_0))}^{(\rho(z),\rho'(z))} \omega_j^{\mathrm{mode}} = \log\frac{\widehat{u}_j(z)}{\widehat{u}_j(z_0)}. \tag{37}$$

The remaining integral follows directly from (17):

$$\int_{z_0}^{z} B_j(\rho(s))\,ds = \sum_{k=1}^{g} c_{jk}\Delta w_k(z). \tag{38}$$

The degree bound is precisely what makes holomorphic differentials sufficient; higher powers require meromorphic differentials as well.

4.4 The mode-solution theorem

Theorem 4.2 (Local hyperelliptic mode solutions)**.** *Let β_j be positive integers, $\sum_{j=1}^N \gamma_j = 0$, and Q a polynomial in independent powers. Define q, r, F by (5)–(7). Assume that F is squarefree of degree $2g+1$ with leading coefficient 4, $g \geq 2$, that the fixed coordinates γ_j/β_j are distinct, and that $q(\gamma_j/\beta_j) \neq 0$. Let $[\rho'(z)]^2 = F(\rho(z))$, $\rho''(z) = F_\rho(\rho(z))/2$ define a nonconstant scalar trajectory on a simply connected local chart with $I_j \neq 0$ and nonvanishing displayed sigma normalizers and point factors. Use compatible infinity-based lifts and (30). Suppose the polynomial corrections (11) satisfy* $\deg B_j \leq g-1$.

Choose initial modes with

$$u_j(z_0)v_j(z_0) = \gamma_j - \beta_j\rho(z_0), \qquad \prod_{j=1}^N u_j(z_0)^{\beta_j} = \frac{q(\rho(z_0)) + \rho'(z_0)}{2}.$$

Then the local canonical solutions are

$$\begin{aligned} u_j(z) &= u_j(z_0)\frac{\widehat{u}_j(z)}{\widehat{u}_j(z_0)} \exp\left[\sum_{k=1}^g c_{jk}\Delta w_k(z)\right], \\ v_j(z) &= v_j(z_0)\frac{\widehat{v}_j(z)}{\widehat{v}_j(z_0)} \exp\left[-\sum_{k=1}^g c_{jk}\Delta w_k(z)\right]. \end{aligned} \tag{39}$$

They satisfy $u_jv_j = I_j$, the product identities (6), $H = 0$, and the weighted invariants (3).

Proof. The point-product identity and initial paired powers give $u_jv_j = I_j$. Differentiating (39), then using (35) and (38), gives

$$\frac{u_j'}{u_j} = \frac{\rho'(\mu_j) - \rho'(z)}{2[\rho(\mu_j) - \rho(z)]} + B_j(\rho(z)) = \frac{\beta_j[q(\rho(z)) - \rho'(z)]}{2I_j(z)} - Q_j(I(z)).$$

This is the u_j equation in (8); the v_j equation follows in the same way. To verify the complete products from the prescribed scalar data, the chain rule gives

$$q_\rho = -\sum_{j=1}^N \beta_j Q_j(I), \qquad r_\rho = -r\sum_{j=1}^N \frac{\beta_j^2}{I_j}.$$

Together with $\rho'' = qq_\rho - 2r_\rho$ and $q^2 - (\rho')^2 = 4r$, these imply

$$\frac{d}{dz}\log\frac{q+\rho'}{2} = \frac{q-\rho'}{2}\sum_{j=1}^N \frac{\beta_j^2}{I_j} - \sum_{j=1}^N \beta_j Q_j(I) = \sum_{j=1}^N \beta_j \frac{u_j'}{u_j}.$$

The initial product condition therefore gives $U = (q+\rho')/2$. Since $UV = r$, it also gives $V = (q-\rho')/2$. Substituting these products into the logarithmic equations recovers the polynomial canonical equations (2). Finally, $H = 0$ follows from $U + V = q$, and the weighted invariants follow from $I_j = \gamma_j - \beta_j\rho$. The canonical vector field is polynomial, so local uniqueness for analytic ordinary differential equations identifies the constructed modes with the solution of the prescribed initial-value problem on this chart. □

The nonzero-power and nonvanishing-factor conditions specify a chart for the logarithmic proof, rather than singularities of the polynomial canonical equations. At a modal zero, those equations can continue a finite solution even though its logarithmic derivative is undefined; the sigma expressions must then be interpreted by continuation or limits. Finite turning points likewise do not obstruct scalar continuation. Continuation around curve cycles is a separate issue and can introduce multipliers, as discussed in Section 8.

Substitution of (34) into (39) gives a sigma ratio normalized at z_0, with coefficients $c_{jk} - h_{jk}$ in the u_j exponential and their negatives in the v_j exponential.

5 Onishi determinants for the complete products

We have already established the Hamiltonian product identities

$$U(z) = \prod_{j=1}^{N} u_j(z)^{\beta_j} = \frac{q(\rho(z)) + \rho'(z)}{2}, \qquad V(z) = \prod_{j=1}^{N} v_j(z)^{\beta_j} = \frac{q(\rho(z)) - \rho'(z)}{2}.$$

Representing these products by Ônishi determinants connects the reconstructed modes to the same addition identities that give their paired powers. The essential step is to account for every elementary mode factor in the determinant's sigma expression, including those contained in its composite factor; this will also explain why the exponential corrections disappear from the complete products.

We know the number N of mode pairs, the genus g of the scalar curve and the polynomial q. We now ask whether the two curve functions on the right can be represented by Ônishi determinants. The size n of the required matrix has yet to be determined. We choose it from the columns available for this genus and the degree of q. We begin with that choice, before relating the determinant's sigma expression to the mode solutions.

5.1 Step 1: determine the columns and size, then fit the product

An Ônishi determinant is formed by evaluating a prescribed list of curve functions at n points, one point per row. The standard column functions are polynomials in the two curve coordinates, ordered by their poles at infinity [16, Theorem 7.2]. They include both pure powers of the coordinate and mixed coordinate–ordinate monomials. The determinant uses the first n functions in this order; we cannot choose an arbitrary collection of columns. Here ρ has pole order two and the monic ordinate $\rho'/2$ has pole order $2g+1$. The beginning of the ordered list is therefore

$$1, \rho, \ldots, \rho^g, \rho'/2, \rho^{g+1}, \rho\rho'/2, \rho^{g+2}, \ldots, \tag{40}$$

with pole orders $0, 2, \ldots, 2g, 2g+1, 2g+2, 2g+3, 2g+4, \ldots$. The factor $1/2$ converts our leading-four curve to Ônishi's monic convention: $(\rho'/2)^2 = F(\rho)/4$, with the same Abel differential $d\rho/[2(\rho'/2)] = d\rho/\rho'$.

To obtain $q \pm \rho'$, the matrix must contain the ordinate column and enough polynomial columns to express the already known q. We choose the shortest initial column list that meets both requirements:

known scalar degree	required columns	chosen size n
$\deg q \le g$	$(1, \rho, \ldots, \rho^g, \rho'/2)$	$g+2$
$\deg q = g+1$	$(1, \rho, \ldots, \rho^g, \rho'/2, \rho^{g+1})$	$g+3$.

These two column arrangements also occur in the interpolation determinants used by Bernatska and Kopeliovich for hyperelliptic divisor reduction [32, Sections 3.1–3.2], up to the ordinate normalization and row order. Here we use them to represent the Hamiltonian products.

Even when $\deg q < g$, we need $g+2$ columns to reach the ordinate. For $\deg q = g+1$, we add the next polynomial column. Thus n is determined here by g and the scalar degree, whereas N was determined by the number of canonical pairs in the Hamiltonian. For the two worked examples presented later in Section 6, these choices give

Known genus g	Known degree $\deg q$	Known mode count N	Chosen size n
2	3	5	5
3	4	7	6

These choices also explain why the determinant contains the ordinate alone, without a polynomial multiplier. There is exactly one ordinate column, $\rho'/2$, and every other column is a power of ρ. Expanding in the moving row gives a constant coefficient of $\rho'/2$, because its cofactor contains only fixed rows. The calculation below identifies that nonzero coefficient as a Vandermonde minor. By contrast, the next column in (40) is $\rho\rho'/2$. Including it would generally give an ordinate term of the form $(a_0 + a_1\rho)\rho'/2$. A restriction with $\deg q > g + 1$ cannot be fitted by the $n = g + 2$ or $n = g + 3$ column arrangements; it requires a longer Ônishi list and a separate argument about these additional ordinate coefficients.

We can now choose $n - 1$ fixed points for the matrix and append the moving point as its last row. The fixed mode points are

$$P_j = (\rho(\mu_j), \rho'(\mu_j)), \qquad \rho'(\mu_j) = q(\rho(\mu_j)),$$

and the moving point is $P(z) = (\rho(z), \rho'(z))$. For the cubic genus-two example, the size just derived is $n = 5$. Evaluating these column functions at the fixed points P_1, P_2, P_3, P_4 gives the first four rows; evaluation at $P(z)$ gives the last row:

$$D_5(z) = \det \begin{pmatrix} 1 & \rho(\mu_1) & \rho(\mu_1)^2 & \rho'(\mu_1)/2 & \rho(\mu_1)^3 \\ 1 & \rho(\mu_2) & \rho(\mu_2)^2 & \rho'(\mu_2)/2 & \rho(\mu_2)^3 \\ 1 & \rho(\mu_3) & \rho(\mu_3)^2 & \rho'(\mu_3)/2 & \rho(\mu_3)^3 \\ 1 & \rho(\mu_4) & \rho(\mu_4)^2 & \rho'(\mu_4)/2 & \rho(\mu_4)^3 \\ 1 & \rho(z) & \rho(z)^2 & \rho'(z)/2 & \rho(z)^3 \end{pmatrix}. \tag{41}$$

Let A denote the indices of the chosen fixed rows; here $A = \{1, 2, 3, 4\}$. Write $q(\rho) = q_0 + q_1\rho + q_2\rho^2 + q_3\rho^3$, and denote the jth matrix column by C_j. Subtracting a linear combination of the other columns from a column leaves the determinant unchanged. We therefore subtract half this polynomial from the fourth column:

$$C_4 \longmapsto C_4 - \tfrac{1}{2}(q_0C_1 + q_1C_2 + q_2C_3 + q_3C_5).$$

Every fixed entry becomes zero because $\rho'(\mu_j) = q(\rho(\mu_j))$. Only the moving entry remains:

$$C_4 \longmapsto \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ [\rho'(z) - q(\rho(z))]/2 \end{pmatrix}.$$

Expansion in this column has sign $(-1)^{5+4} = -1$ and leaves the fixed-point Vandermonde minor. Hence

$$D_5(z) = \Delta_A \frac{q(\rho(z)) - \rho'(z)}{2}, \qquad \Delta_A = \prod_{1 \le i < j \le 4} [\rho(\mu_j) - \rho(\mu_i)].$$

We have made the determinant reproduce V directly. Let ι (the Greek letter iota) denote the sheet involution, which reverses the ordinate while preserving the coordinate: $\iota(P_j) = (\rho(\mu_j), -\rho'(\mu_j))$ and write $P_j^* = \iota(P_j)$. Here the star denotes the sheet involution, not complex conjugation. Reversing the ordinates of the fixed points, that is $\rho'(\mu_j) \mapsto -\rho'(\mu_j)$ for $j \in A$, while leaving the moving point unchanged, replaces their polynomial values by $-q$. The same operation then gives $D_5^*(z) = -\Delta_A[q(\rho(z)) + \rho'(z)]/2$, reproducing U.

The same column operation works in any genus for $n = g+2$ with $\deg q \le g$, or for $n = g+3$ with $\deg q = g + 1$. The polynomial q is already specified; its fixed-point values allow us to subtract it from the ordinate column. Distinct fixed coordinates ensure that the remaining Vandermonde minor is nonzero. Thus the column operation expresses the determinant as a constant multiple of the complete mode product $V = (q - \rho')/2$.

Lemma 5.1. *Let A index $n-1$ chosen fixed points, and form D_n by evaluating the stated column functions at these points and at $P(z)$. Let D_n^* instead evaluate its fixed rows at $P_j^* = \iota(P_j)$, leaving the moving point unchanged. Define*

$$\Delta_A = \prod_{\substack{i,j\in A\\ i<j}} [\rho(\mu_j) - \rho(\mu_i)], \qquad \varepsilon = \begin{cases} 1, & n = g+2, \\ -1, & n = g+3. \end{cases}$$

Then

$$D_n(z) = \varepsilon\Delta_A \frac{\rho'(z) - q(\rho(z))}{2}, \qquad D_n^*(z) = \varepsilon\Delta_A \frac{\rho'(z) + q(\rho(z))}{2}. \tag{42}$$

Proof. Subtract $q/2$ from the ordinate column and expand its sole nonzero entry, as above. The ordinate is the last column when $n = g+2$; it precedes the last polynomial column when $n = g+3$, giving the sign ε. For the fixed points on the opposite sheet subtract $-q/2$ instead. □

5.2 The Schur complement and the Vandermonde inverse

The same determinant reduction can be expressed as a Schur complement. This makes the role of interpolation explicit: the fixed ordinates determine the polynomial part of the moving row through the inverse of the fixed-coordinate matrix.

First move the ordinate column to the last position. For $n = g+2$ it is already there; for $n = g+3$ this requires one column interchange. The resulting matrix has the block form

$$\mathcal{M}_n(z) = \begin{pmatrix} \mathsf{W}_A & \boldsymbol{\rho}'_A/2 \\ \mathbf{p}(\rho(z))^T & \rho'(z)/2 \end{pmatrix}, \qquad \det \mathcal{M}_n(z) = \varepsilon D_n(z), \tag{43}$$

where, in the order of the fixed rows,

$$\mathbf{p}(\rho) = (1, \rho, \dots, \rho^{n-2})^T, \qquad \mathsf{W}_A = \big(\rho(\mu_j)^k\big)_{j\in A,\, 0\le k\le n-2}, \qquad \boldsymbol{\rho}'_A = \big(\rho'(\mu_j)\big)_{j\in A}.$$

Here $\boldsymbol{\rho}'_A$ is a column vector, and W_A is the $(n-1)\times(n-1)$ Vandermonde matrix with determinant Δ_A. It is invertible because the selected fixed coordinates are distinct.

If $\mathbf{q} = (q_0, \dots, q_{n-2})^T$ is the coefficient vector of the prescribed q, padded with zeros when necessary, its fixed-point values satisfy

$$\mathsf{W}_A \mathbf{q} = \boldsymbol{\rho}'_A.$$

Consequently,

$$q(\rho) = \mathbf{p}(\rho)^T \mathsf{W}_A^{-1} \boldsymbol{\rho}'_A. \tag{44}$$

Thus the Vandermonde inverse takes the fixed ordinates $\rho'(\mu_j)$ to the polynomial q evaluated at the moving coordinate. This is interpolation of the already specified scalar restriction, expressed in matrix form.

The Schur complement of W_A in (43) is

$$\mathcal{S}(z) = \frac{\rho'(z)}{2} - \mathbf{p}(\rho(z))^T \mathsf{W}_A^{-1} \frac{\boldsymbol{\rho}'_A}{2} = \frac{\rho'(z) - q(\rho(z))}{2}. \tag{45}$$

Block elimination gives $\det \mathcal{M}_n = \det \mathsf{W}_A \, \mathcal{S}$, and hence

$$\frac{D_n(z)}{\Delta_A} = \varepsilon \mathcal{S}(z).$$

For the later worked examples $\varepsilon = -1$, so $D_n/\Delta_A = -\mathcal{S} = V$. Applying the sheet involution only to the fixed points negates $\boldsymbol{\rho}'_A$ and leaves W_A unchanged. The corresponding Schur complement is $[\rho'(z) + q(\rho(z))]/2 = U$, so $U = \varepsilon D_n^*/\Delta_A$. In the original $n = g+3$ column order, the upper-left block is not the Vandermonde matrix; the column interchange and its sign are essential to this interpretation.

5.3 Step 2: relate the sigma factors to the modes

Before counting the mode factors, we display the sigma side of the identity. For the genus-two five-row determinant (41), Ônishi's formula [16, Theorem 7.2], with its corrected sign, reads

$$D_5(z) = -\frac{\sigma\left(w(z)+\sum_{i=1}^{4} w(\mu_i)\right)\prod_{1\le i<j\le 4}\sigma(w(\mu_i)-w(\mu_j))\prod_{i=1}^{4}\sigma(w(\mu_i)-w(z))}{\sigma_2(w(z))^5\prod_{i=1}^{4}\sigma_2(w(\mu_i))^5}. \tag{46}$$

The right side contains two kinds of moving sigma arguments: differences with individual fixed points, and a sum involving all four fixed points. The fixed–fixed differences are constants. Grouping the four individual moving factors with their normalizers gives the factors $\bar{v}_i$ defined in Lemma 4.1; we call these *elementary point factors* because each is attached to one fixed point. What remains, apart from a fixed constant, is

$$\frac{\sigma(w(z)+w(\mu_1)+\cdots+w(\mu_4))}{\sigma_2(w(z))}.$$

We call this the *composite factor* because its shift combines the fixed Abel vectors. We will show that it supplies the additional elementary factors required by the complete mode product. The same grouping applies to the two determinant families above in any genus.

The determinant vanishes when the moving point satisfies $\rho' = q(\rho)$. The identity

$$(q-\rho')(q+\rho') = 4r(\rho) = 4\prod_{j=1}^{N}[\gamma_j-\beta_j\rho]^{\beta_j}$$

locates every such zero, including its multiplicity. At P_j, the other factor is $q+\rho' = 2q(\rho(\mu_j)) \neq 0$, so the zero of $q-\rho'$ has order exactly β_j. Their total multiplicity is $M = \sum_{j=1}^{N}\beta_j$. This count is algebraic: with the fixed rows held constant, the determinant is proportional to the meromorphic curve function $q(\rho)-y$. We count its zeros and poles on the compact curve, not in the propagation variable z or on the full Jacobian. Its only pole is at infinity; in a local curve parameter t there, $\rho = t^{-2}$ and $y \sim \pm 2t^{-(2g+1)}$, so their pole orders are 2 and $2g+1$. If $\deg q \le g$, the y term gives pole order $2g+1$; if $\deg q = g+1$, the leading term of q gives pole order $2g+2$. A meromorphic function has equal total zero and pole orders, so $M = 2g+1$ or $2g+2$, respectively. Removing the $n-1$ occurrences represented by the fixed rows leaves $M-(n-1) = g$ further occurrences. These are the remaining intersections of $y = q(\rho)$ with the curve, called the complementary intersection divisor in [32, Sections 3.1–3.2]. The count identifies the residual points but does not yet prove their sigma factorization. Ônishi's identity applies with the moving argument constrained to the Abel image of this same curve; below, an auxiliary determinant and the two-point identity establish how the composite factor supplies the residual elementary factors. No global meromorphicity in z, or of each individual sigma factor, is required.

In the genus-two example the complete list is $P_1, P_1, P_2, P_3, P_4, P_5$. The repetition of P_1 records $\beta_1 = 2$; in general a point occurs more than once precisely when $\beta_j > 1$. The fixed rows of (41) account for P_1, P_2, P_3, P_4, leaving P_1, P_5. The determinant's sigma expression must account for these two further point factors, one for mode 1 and one for mode 5. We now identify where they occur and prove their factorization.

Let B index the residual points not yet accounted for by the fixed rows indexed by A. We choose the fixed and residual points so each group has distinct coordinates internally, allowing a point to belong to both groups. Thus their combined occurrences reproduce the weights. For the two examples,

Genus	Fixed points indexed by A	Residual points indexed by B
2	P_1, P_2, P_3, P_4	P_1, P_5
3	P_1, P_2, P_3, P_4, P_5	P_1, P_6, P_7

There is no repeated row inside either determinant. The repeated factor will appear when the two lists are combined.

The complete zero list also fixes the summed Abel argument. Abel's theorem says that the weighted sum of the Abel vectors of the zeros of $q - \rho'$, whose poles are all at infinity, is a period vector. Choose their integration paths so that this sum is zero, and put

$$\sum_{j=1}^{N} \beta_j w(\mu_j) = 0, \qquad E = \sum_{j\in A} w(\mu_j) = -\sum_{j\in B} w(\mu_j).$$

Both examples have unit-weight points, so the period vector can be absorbed into one such point's path. Sigma normalizers and h_{jk} must use these same fixed lifts. The residual points thus have the opposite Abel sum to the chosen fixed points. Applying the sheet involution to the residual points negates their Abel sum, making it equal to that of the chosen points modulo periods. This is the divisor-class relation used in interpolation-based reduction [32]; here it explains the opposite-sheet points in the auxiliary determinant below.

Ônishi's identity [16, Theorem 7.2] now groups into

$$D_n(z) = K_A \frac{\sigma(w(z) + E)}{\sigma_\sharp(w(z))} \prod_{j\in A} \bar{v}_j(z), \tag{47}$$

Here K_A is a nonzero scalar constant formed from the fixed-point sigma values and normalizers in the identity; it is independent of z. The $n-1$ factors $\bar{v}_j$ are the familiar elementary factors of Lemma 4.1. The additional ratio $\sigma(w+E)/\sigma_\sharp(w)$ has a summed fixed argument, and is the composite factor we must resolve. The fixed constants use the sign corrected by Ônishi in his published errata to the determinant identity, $c_n = (-1)^{(n-g-1)(n+g^2+2g)/2}$ for $n \geq g$.

For genus two, apply the sheet involution to the residual points, using $P_1^* = \iota(P_1), P_5^* = \iota(P_5)$ and the moving point in an auxiliary three-row identity. Its columns are purely polynomial, so

$$\det \begin{pmatrix} 1 & \rho(\mu_1) & \rho(\mu_1)^2 \\ 1 & \rho(\mu_5) & \rho(\mu_5)^2 \\ 1 & \rho(z) & \rho(z)^2 \end{pmatrix} = \Delta_B[\rho(z) - \rho(\mu_1)][\rho(z) - \rho(\mu_5)], \tag{48}$$

where $\Delta_B = \rho(\mu_5) - \rho(\mu_1)$. The sheet involution leaves these polynomial entries unchanged, but gives the summed Abel argument $w - w(\mu_1) - w(\mu_5) = w + E$ in its sigma expression. Its other moving factors are $\bar{u}_1\bar{u}_5$. The full equality supplied by the sigma identity is

$$\det \begin{pmatrix} 1 & \rho(\mu_1) & \rho(\mu_1)^2 \\ 1 & \rho(\mu_5) & \rho(\mu_5)^2 \\ 1 & \rho(z) & \rho(z)^2 \end{pmatrix} = K_B \frac{\sigma(w+E)}{\sigma_2(w)} \bar{u}_1\bar{u}_5.$$

Here K_B is another nonzero scalar constant determined by the fixed residual points, independent of z. On the other hand, the two-point identity gives $\bar{u}_j\bar{v}_j = \rho - \rho(\mu_j)$ in genus two. Thus (48) is also $\Delta_B\bar{u}_1\bar{v}_1\bar{u}_5\bar{v}_5$. Comparing these expressions and cancelling $\bar{u}_1\bar{u}_5$ proves

$$\frac{\sigma(w+E)}{\sigma_2(w)} = \Gamma_B\bar{v}_1\bar{v}_5, \qquad \Gamma_B = \Delta_B/K_B.$$

Substitution into (47) gives $D_5 = K\bar{v}_1^2\bar{v}_2\bar{v}_3\bar{v}_4\bar{v}_5$, with $K = K_A\Gamma_B$. The composite factor has supplied exactly the missing occurrences: it is not an unexplained extra mode.

In general the auxiliary determinant has the g residual points on the opposite sheet and the moving point, with columns $1, \rho, \dots, \rho^g$. Its Vandermonde expression contains $\prod_{j\in B}[\rho-\rho(\mu_j)]$; its sigma expression contains $[\sigma(w+E)/\sigma_\sharp(w)]\prod_{j\in B}\bar{u}_j$. Comparing them through (33) proves, by the same cancellation,

$$\frac{\sigma(w+E)}{\sigma_\sharp(w)} = \Gamma_B \prod_{j\in B} \bar{v}_j, \qquad \Gamma_B \neq 0 \text{ constant in } z. \tag{49}$$

We cancel on a regular chart and extend the resulting identity wherever the continued expressions are defined. This is a factorization along the curve's Abel image, not on the entire Jacobian. Combining the main and residual factors, and then multiplying the two-point identities, yields

$$\prod_{j=1}^{N} \bar{u}_j^{\beta_j} = K_+[q(\rho)+\rho'], \qquad \prod_{j=1}^{N} \bar{v}_j^{\beta_j} = K_-[q(\rho)-\rho'], \tag{50}$$

where K_+, K_- are nonzero fixed constants. All the weighted elementary factors have now been recovered from the determinant identity.

5.4 Step 3: deduce cancellation of the exponential corrections

The cancellation is a consequence of the complete factorization established in Step 2 together with the differential identities for the modes. We now derive that consequence explicitly. The full modes include both the point factors and exponential corrections. Multiplying their normalized sigma expressions gives

$$U(z) = U(z_0)\prod_{j=1}^{N}\left[\frac{\bar{u}_j(z)}{\bar{u}_j(z_0)}\right]^{\beta_j} \exp\left[\sum_{k=1}^{g}\left\{\sum_{j=1}^{N}\beta_j(c_{jk}-h_{jk})\right\}\Delta w_k\right]. \tag{51}$$

The corresponding V formula uses $\bar{v}_j$ and the negative exponent. We must therefore show that the weighted coefficients in braces vanish.

Step 2 identifies $\prod_{j=1}^{N}\bar{u}_j^{\beta_j}$ with a constant times $q+\rho'$. Their logarithmic derivatives are equal. Using the kernel differential on one side and the scalar chain rule on the other gives two expressions for this same derivative:

$$\frac{d}{dz}\log\prod_{j=1}^{N}\bar{u}_j^{\beta_j} = \sum_{j=1}^{N}\beta_j\frac{\rho'(\mu_j)-\rho'}{2[\rho(\mu_j)-\rho]} + \sum_{k=1}^{g}\left(\sum_{j=1}^{N}\beta_j h_{jk}\right)\rho^{k-1},$$

$$\frac{d}{dz}\log(q+\rho') = \sum_{j=1}^{N}\beta_j\frac{\rho'(\mu_j)-\rho'}{2[\rho(\mu_j)-\rho]} + \sum_{k=1}^{g}\left(\sum_{j=1}^{N}\beta_j c_{jk}\right)\rho^{k-1}.$$

The rational point terms are identical and cancel on subtraction. The remaining polynomial vanishes for every z in the regular local domain. Since $\rho(z)$ is nonconstant, its values contain infinitely many distinct coordinates with an accumulation point, so this polynomial is identically zero. Each coefficient therefore vanishes:

$$\sum_{j=1}^{N}\beta_j(c_{jk}-h_{jk}) = 0, \qquad k = 1, \dots, g. \tag{52}$$

This is why the exponential in (51) becomes one; it is a consequence of the complete Ônishi factorization, not a separate normalization imposed on the individual modes.

Only the normalized point-factor products remain. Equation (50) gives

$$U(z) = U(z_0)\frac{q(\rho(z)) + \rho'(z)}{q(\rho(z_0)) + \rho'(z_0)}, \qquad V(z) = V(z_0)\frac{q(\rho(z)) - \rho'(z)}{q(\rho(z_0)) - \rho'(z_0)}.$$

The initial product conditions fix these as $(q + \rho')/2$ and $(q - \rho')/2$. Step 1 then identifies them with the normalized determinants:

$$U(z) = \varepsilon\frac{D_n^*(z)}{\Delta_A}, \qquad V(z) = -\varepsilon\frac{D_n(z)}{\Delta_A}. \tag{53}$$

For both worked examples $\varepsilon = -1$, hence $U = -D_n^*/\Delta_A$ and $V = D_n/\Delta_A$. This completes the connection between the Hamiltonian products, the elementary sigma solutions and Ônishi's determinant.

The balanced fixed lifts make the cancellation visible in the explicit exponent. Other compatible lifts can redistribute an exponential between the raw sigma product and its correction, but the normalized initial-value solutions, and therefore their determinant products, are unchanged. The construction here uses the two stated polynomial column spaces and internally distinct main and residual lists, which both examples satisfy. Higher scalar degree requires longer column lists; repeated rows within a list require derivative limits rather than duplicating a row. Bernatska and Kopeliovich treat the corresponding rational interpolation and repeated-point constructions in [32, Sections 3.3–3.5]. Extending the present sigma-factor proof to those cases is outside our scope.

6 Worked examples

We now apply the construction to two explicit Hamiltonian systems and give their analytic mode solutions: a five-pair genus-two example and a seven-pair genus-three example.

6.1 Genus-two example with repeated mode

We construct a five-pair system on a smooth genus-two curve, including a double mode and affine polynomial corrections.

6.1.1 Polynomial data

Choose degeneracies and power offsets:

$$\beta = (2, 1, 1, 1, 1), \qquad \gamma = \left(-\frac{3}{5}, -\frac{1}{5}, -\frac{1}{10}, \frac{2}{5}, \frac{1}{2}\right). \tag{54}$$

Note $\sum_{j=1}^5 \beta_j = 6$ and $\sum_{j=1}^5 \gamma_j = 0$. The centred power convention gives

$$\rho = -\frac{\sum_{j=1}^5 x_j}{\sum_{j=1}^5 \beta_j} = -\frac{x_1 + x_2 + x_3 + x_4 + x_5}{6}.$$

Define the polynomial $Q(x_1, \ldots, x_5)$ in independent power variables:

$$\begin{aligned} Q(x_1, \ldots, x_5) = 4&\left(-\frac{S(x)}{6} + \frac{27}{40}\right)\left(-\frac{S(x)}{6} + \frac{1}{20}\right)\left(-\frac{S(x)}{6} - \frac{3}{5}\right) \\ &+ \frac{1}{3}\left(\frac{x_1}{2} - x_2 + \frac{1}{10}\right)x_3, \qquad S(x) = \sum_{k=1}^5 x_k. \end{aligned} \tag{55}$$

The second term is quadratic and vanishes on the power line, but changes individual partial derivatives. Writing the first term as $q(-S(x)/6)$, the full derivatives are

$$\begin{aligned} Q_1 &= -q_\rho(-S(x)/6)/6 + x_3/6, & Q_2 &= -q_\rho(-S(x)/6)/6 - x_3/3, \\ Q_3 &= -q_\rho(-S(x)/6)/6 + (x_1/2 - x_2 + 1/10)/3, & Q_4 &= Q_5 = -q_\rho(-S(x)/6)/6. \end{aligned}$$

One must differentiate the polynomial in the independent powers before restricting to the line: q alone does not specify the individual modes.

6.1.2 The explicit coupled equations

For the full system, set $I(z) = (u_1(z)v_1(z), \dots, u_5(z)v_5(z))$. The polynomial multiplying each u_j is $\Phi_j^{(2)}(I) = -Q_j(I)$; the same polynomial enters the v_j equation with the opposite sign. It is useful to display these terms before restricting to the scalar trajectory. Define

$$A_2(x) = \frac{S(x)^2}{18} - \frac{S(x)}{36} - \frac{107}{400}, \qquad S(x) = \sum_{k=1}^{5} x_k. \tag{56}$$

Then

$$\begin{aligned} \Phi_1^{(2)}(x) &= A_2(x) - x_3/6, \\ \Phi_2^{(2)}(x) &= A_2(x) + x_3/3, \\ \Phi_3^{(2)}(x) &= A_2(x) - (x_1/2 - x_2 + 1/10)/3, \\ \Phi_j^{(2)}(x) &= A_2(x), \qquad j = 4, 5. \end{aligned} \tag{57}$$

The coupled polynomial equations are

$$\begin{aligned} u_1' &= \Phi_1^{(2)}(I)u_1 + 2v_1 \prod_{k=2}^{5} v_k, \\ v_1' &= -\Phi_1^{(2)}(I)v_1 - 2u_1 \prod_{k=2}^{5} u_k, \\ u_j' &= \Phi_j^{(2)}(I)u_j + v_1^2 \prod_{\substack{2 \le k \le 5 \\ k \ne j}} v_k, \\ v_j' &= -\Phi_j^{(2)}(I)v_j - u_1^2 \prod_{\substack{2 \le k \le 5 \\ k \ne j}} u_k, \qquad j = 2, \dots, 5. \end{aligned} \tag{58}$$

Every multiplicative term is thus an explicit polynomial in the paired products $u_k v_k$. The separate first equation pair shows the factor 2 and the extra mode factor required by the double weight. The remaining equations have the unit-weight coupling products.

6.1.3 Restriction to the power line

On the power line $x_j = \gamma_j - \beta_j \rho$, we substitute to obtain the restricted polynomials. Define

$$Q(\gamma_1 - \beta_1\rho, \dots, \gamma_5 - \beta_5\rho) = q(\rho), \qquad \prod_{j=1}^{5} (\gamma_j - \beta_j\rho)^{\beta_j} = r(\rho). \tag{59}$$

Explicitly, the restriction gives

$$q(\rho) = 4\left(\rho + \frac{27}{40}\right)\left(\rho + \frac{1}{20}\right)\left(\rho - \frac{3}{5}\right). \tag{60}$$

The weighted power product is

$$r(\rho) = 4\left(\rho + \frac{3}{10}\right)^2\left(\rho + \frac{1}{5}\right)\left(\rho + \frac{1}{10}\right)\left(\rho - \frac{2}{5}\right)\left(\rho - \frac{1}{2}\right). \tag{61}$$

The scalar polynomial $F(\rho) = q(\rho)^2 - 4r(\rho)$ has degree-six terms that cancel, leaving

$$\begin{aligned} F(\rho) = 4\rho^5 - \frac{747\rho^4}{100} - \frac{1581\rho^3}{1000} + \frac{83993\rho^2}{40000} \\ + \frac{16113\rho}{100000} + \frac{801}{1000000}. \end{aligned} \tag{62}$$

This is squarefree of degree five with leading coefficient 4, as required for a smooth genus-two curve.

6.1.4 Polynomial corrections

The fixed mode points satisfy $\rho(\mu_j) = \gamma_j/\beta_j$, giving

$$(\rho(\mu_1), \ldots, \rho(\mu_5)) = \left(-\frac{3}{10}, -\frac{1}{5}, -\frac{1}{10}, \frac{2}{5}, \frac{1}{2}\right).$$

Exact evaluation of (11) gives

$$B_1(\rho) = \frac{41\rho}{60} + \frac{67}{150}, \tag{63}$$

$$B_2(\rho) = \frac{283}{600} - \frac{\rho}{60}, \tag{64}$$

$$B_3(\rho) = \frac{7\rho}{60} + \frac{27}{50}, \tag{65}$$

$$B_4(\rho) = \frac{23}{200} - \frac{53\rho}{60}, \tag{66}$$

$$B_5(\rho) = -\frac{13\rho}{12} - \frac{9}{100}, \tag{67}$$

All corrections are affine, with unequal linear coefficients. The quadratic terms cancel between the divided difference and the full partial of Q, although Q itself is cubic. Thus $\deg B_j = 1 < g = 2$ without a rescaling of the curve. The coefficients c_{j1}, c_{j2} in the mode formulas are the constant and linear coefficients displayed above.

6.1.5 Mode solutions

The Abel integrals are

$$w_1(z) = \int_\infty^{(\rho(z),\rho'(z))} \frac{d\rho}{y}, \qquad w_2(z) = \int_\infty^{(\rho(z),\rho'(z))} \frac{\rho\, d\rho}{y}.$$

The scalar inversion is

$$\rho(z) = -\frac{\sigma_1(w(z))}{\sigma_2(w(z))}, \qquad \sigma(w(z)) = 0.$$

The point factors are given by (34) with $g = 2$:

$$\widehat{u}_j(z) = \sqrt{\beta_j}\,\frac{\sigma(w(z) + w(\mu_j))}{\sigma_2(w(z))\sigma_2(w(\mu_j))}\exp[-h_{j1}\Delta w_1(z) - h_{j2}\Delta w_2(z)]\,,$$

where $\Delta w_k(z) = w_k(z) - w_k(z_0)$ and the constants h_{j1}, h_{j2} are given by (31).

The complete mode solutions are

$$\begin{aligned} u_j(z) &= u_j(z_0)\frac{\widehat{u}_j(z)}{\widehat{u}_j(z_0)}\exp[c_{j1}\Delta w_1(z) + c_{j2}\Delta w_2(z)]\,, \\ v_j(z) &= v_j(z_0)\frac{\widehat{v}_j(z)}{\widehat{v}_j(z_0)}\exp[-c_{j1}\Delta w_1(z) - c_{j2}\Delta w_2(z)]\,. \end{aligned} \tag{68}$$

These solve the canonical system for Hamiltonian

$$H = \prod_{j=1}^{5} u_j^{\beta_j} + \prod_{j=1}^{5} v_j^{\beta_j} - Q(u_1v_1, \ldots, u_5v_5),$$

with $H = 0$ and $I_j - \beta_j I_5 = \gamma_j - \beta_j\gamma_5$ for $j < 5$.

6.2 Genus-three example with quadratic corrections

We now construct a seven-pair system on a smooth genus-three curve with quadratic polynomial corrections.

6.2.1 Polynomial data

Choose degeneracies and power offsets:

$$\beta = (2,1,1,1,1,1,1), \qquad \gamma = \left(-\frac{3}{5}, -\frac{1}{4}, -\frac{1}{5}, -\frac{1}{10}, \frac{1}{5}, \frac{2}{5}, \frac{11}{20}\right). \tag{69}$$

Note $\sum_{j=1}^{7}\beta_j = 8$ and $\sum_{j=1}^{7}\gamma_j = 0$. The centred power convention gives

$$\rho = -\frac{\sum_{j=1}^{7} x_j}{\sum_{j=1}^{7}\beta_j} = -\frac{x_1 + \cdots + x_7}{8}.$$

Define the polynomial $Q(x_1, \ldots, x_7)$ in independent power variables:

$$\begin{aligned} Q(x_1, \ldots, x_7) = 4&\left(-\frac{S(x)}{8} + \frac{37}{40}\right)\left(-\frac{S(x)}{8} + \frac{3}{20}\right)\left(-\frac{S(x)}{8} - \frac{3}{10}\right)\left(-\frac{S(x)}{8} - \frac{13}{20}\right) \\ &+ \frac{1}{7}\left(\frac{x_1}{2} - x_2 + \frac{1}{20}\right)x_3^2, \qquad S(x) = \sum_{k=1}^{7} x_k. \end{aligned} \tag{70}$$

The first term depends only on $\sum_{j=1}^{7} x_j$. The second term vanishes on the power line $x_j = \gamma_j - \beta_j\rho$ but has nonzero partials.

6.2.2 The explicit coupled equations

For the full system, set $I(z) = (u_1(z)v_1(z), \ldots, u_7(z)v_7(z))$. The polynomial multiplying each u_j is $\Phi_j^{(3)}(I) = -Q_j(I)$; the same polynomial enters the v_j equation with the opposite sign. It is useful to display these terms before restricting to the scalar trajectory. Define

$$A_3(x) = -\frac{S(x)^3}{256} + \frac{3S(x)^2}{1024} + \frac{11S(x)}{128} + \frac{249}{6400}, \qquad S(x) = \sum_{k=1}^{7} x_k. \tag{71}$$

Then

$$\begin{aligned}
\Phi_1^{(3)}(x) &= A_3(x) - x_3^2/14,\\
\Phi_2^{(3)}(x) &= A_3(x) + x_3^2/7,\\
\Phi_3^{(3)}(x) &= A_3(x) - 2(x_1/2 - x_2 + 1/20)x_3/7,\\
\Phi_j^{(3)}(x) &= A_3(x), \qquad j = 4, \dots, 7.
\end{aligned} \tag{72}$$

The coupled polynomial equations are

$$\begin{aligned}
u_1' &= \Phi_1^{(3)}(I)u_1 + 2v_1 \prod_{k=2}^{7} v_k,\\
v_1' &= -\Phi_1^{(3)}(I)v_1 - 2u_1 \prod_{k=2}^{7} u_k,\\
u_j' &= \Phi_j^{(3)}(I)u_j + v_1^2 \prod_{\substack{2\le k\le 7\\ k\neq j}} v_k,\\
v_j' &= -\Phi_j^{(3)}(I)v_j - u_1^2 \prod_{\substack{2\le k\le 7\\ k\neq j}} u_k, \qquad j = 2, \dots, 7.
\end{aligned} \tag{73}$$

Every multiplicative term is thus an explicit polynomial in the paired products $u_k v_k$. The separate first equation pair shows the factor 2 and the extra mode factor required by the double weight. The remaining equations have the unit-weight coupling products.

6.2.3 Restriction to the power line

On the power line $x_j = \gamma_j - \beta_j \rho$, the restriction gives

$$q(\rho) = 4\left(\rho + \frac{37}{40}\right)\left(\rho + \frac{3}{20}\right)\left(\rho - \frac{3}{10}\right)\left(\rho - \frac{13}{20}\right). \tag{74}$$

The weighted power product is

$$\begin{aligned}
r(\rho) = 4&\left(\rho + \frac{3}{10}\right)^2\left(\rho + \frac{1}{4}\right)\left(\rho + \frac{1}{5}\right)\left(\rho + \frac{1}{10}\right)\\
&\times\left(\rho - \frac{1}{5}\right)\left(\rho - \frac{2}{5}\right)\left(\rho - \frac{11}{20}\right).
\end{aligned} \tag{75}$$

The degree-eight terms cancel in $F = q^2 - 4r$, giving

$$\begin{aligned}
F(\rho) = 4\rho^7 &- \frac{1539\rho^6}{100} + \frac{74\rho^5}{125} + \frac{160751\rho^4}{20000}\\
&- \frac{176721\rho^3}{100000} - \frac{7797647\rho^2}{16000000} + \frac{5803941\rho}{80000000} + \frac{19247121}{1600000000}.
\end{aligned} \tag{76}$$

Exact squarefreeness gives a smooth genus-three curve with leading coefficient 4.

6.2.4 Polynomial corrections

Write $B_j(\rho) = c_{j1} + c_{j2}\rho + c_{j3}\rho^2$. The exact coefficients are listed in Table 1. The cubic terms cancel, leaving $\deg B_j = 2 < g = 3$, although Q is quartic. All three coordinates contribute through $\sum_{k=1}^{3} c_{jk}\Delta w_k$. As in genus two, the term vanishing on the power line changes the individual corrections without changing q or the scalar curve.

j	c_{j1}	c_{j2}	c_{j3}
1	$-\frac{112129}{224000}$	$\frac{1551}{2800}$	$\frac{261}{560}$
2	$-\frac{19673}{44800}$	$\frac{191}{280}$	$\frac{65}{112}$
3	$-\frac{12343}{32000}$	$\frac{263}{400}$	$\frac{27}{80}$
4	$-\frac{8151}{32000}$	$\frac{277}{400}$	$\frac{11}{80}$
5	$\frac{4233}{32000}$	$\frac{223}{400}$	$-\frac{37}{80}$
6	$\frac{8489}{32000}$	$\frac{107}{400}$	$-\frac{69}{80}$
7	$\frac{7397}{32000}$	$-\frac{11}{200}$	$-\frac{93}{80}$

Table 1: Genus-three polynomial corrections, generated by restriction of the full partial derivatives of Q.

6.2.5 Mode solutions

The Abel integrals are

$$w_k(z) = \int_{\infty}^{(\rho(z),\rho'(z))} \frac{\rho^{k-1} d\rho}{y}, \qquad k = 1, 2, 3.$$

The scalar inversion is

$$\rho(z) = -\frac{\sigma_{13}(w(z))}{\sigma_{23}(w(z))}, \qquad \sigma(w(z)) = \sigma_3(w(z)) = 0.$$

The point factors are given by (34) with $g = 3$:

$$\widehat{u}_j(z) = \sqrt{\beta_j} \frac{\sigma_3(w(z) + w(\mu_j))}{\sigma_2(w(z))\sigma_2(w(\mu_j))} \exp\left[-\sum_{k=1}^{3} h_{jk} \Delta w_k(z) \right],$$

where the constants h_{j1}, h_{j2}, h_{j3} are given by (31).

The complete mode solutions are

$$\begin{aligned} u_j(z) &= u_j(z_0) \frac{\widehat{u}_j(z)}{\widehat{u}_j(z_0)} \exp\left[\sum_{k=1}^{3} c_{jk} \Delta w_k(z) \right], \\ v_j(z) &= v_j(z_0) \frac{\widehat{v}_j(z)}{\widehat{v}_j(z_0)} \exp\left[-\sum_{k=1}^{3} c_{jk} \Delta w_k(z) \right]. \end{aligned} \tag{77}$$

These solve the canonical system for the seven-pair Hamiltonian with quadratic mode corrections.

7 Numerical validation

We compare the sigma solutions with independent numerical integration of the full polynomial canonical equations, first on an exterior genus-two branch and then over three oscillations of a bounded genus-three branch. Numerous symbolic and numerical checks were carried out in Python; the code is provided in the repository identified in the code availability statement.

Both comparisons use 30-digit working precision. This specifies the arithmetic, not the achieved accuracy: the observed discrepancies and their reduction under RK4 step halving

provide the accuracy evidence. Let s denote the state vector containing all u_j and v_j modes. The scaled mode discrepancy at an observation time is

$$E_{\mathrm{mode}} = \max_{\ell} \frac{|s_\ell^{\mathrm{RK4}} - s_\ell^{\sigma}|}{\max(1, |s_\ell^{\sigma}|)}.$$

The sigma solution is sampled on a sparse grid, while RK4 uses a finer grid containing those observation times. Step halving distinguishes finite-step integration error from arithmetic precision. At each observation time, numerical inversion of the scalar Abelian integral determines $\rho(z)$. We then evaluate the Abel vector and the prescribed sigma formulas for the individual modes; the scalar sigma ratio provides a consistency check. The polynomial and exponential coefficients are prescribed analytically, without fitting to trajectory data.

Table 2 reports the existing sigma-based consistency checks separately from the RK4 comparisons. The scalar sigma ratio is compared with the quadrature-inverted coordinate; paired powers and complete products are compared with their prescribed curve expressions. Coordinate discrepancies are divided by $\max(1, |\rho|)$, paired-power discrepancies by $\max(1, |I_j^{\sigma}|, |\gamma_j - \beta_j \rho|)$, and complete-product discrepancies by $\max(1, |U^{\sigma}|)$ or $\max(1, |V^{\sigma}|)$. Here the superscript σ denotes evaluation from the sigma mode formulas. The genus-three coordinate lies below one in magnitude, so its absolute and scaled coordinate discrepancies coincide. These checks test the coordinate representation and algebraic identities at the computed Abel vectors; they do not replace scalar inversion at prescribed propagation or provide certified bounds on all sigma-evaluation errors.

Maximum scaled discrepancy	Genus two	Genus three
Scalar sigma coordinate	5.672682×10^{-31}	9.423342×10^{-30}
Paired modal powers	8.376260×10^{-31}	5.994113×10^{-29}
Complete product U	1.461370×10^{-30}	8.992481×10^{-30}
Complete product V	2.264083×10^{-30}	2.818472×10^{-29}

Table 2: Sigma-based consistency checks at 30-digit working precision, over 33 genus-two observations and 97 uniform genus-three observations plus 18 turning-point and modal-zero observations. Values are maxima from the supplied numerical reports.

7.1 Extended genus-two trajectory

For the first example, we propagate from $\rho(z_0) = 4.5$ to $\rho = 10$ on the positive sheet. The propagation length is $T = 0.02960302978454372$. We use the Hamiltonian, weights and invariant offsets of the first worked example, with initial modes in the proportions

$$(u_1(z_0), \ldots, u_5(z_0)) \;\propto\; (1.1 + 0.2i,\; 0.8 - 0.1i,\; 1.2 + 0.1i,\; 1 + 0.1i,\; 1.1 + 0.1i).$$

Their common factor is the principal sixth root that enforces

$$u_1(z_0)^2 \prod_{j=2}^{5} u_j(z_0) = \frac{q(4.5) + \sqrt{F(4.5)}}{2}, \qquad v_j(z_0) = \frac{\gamma_j - 4.5\beta_j}{u_j(z_0)}.$$

This specifies the initial state on the prescribed invariant level.

The largest real branch point is approximately 1.92552416. Our exterior branch therefore has $\rho'(z) = \sqrt{F(\rho(z))} > 0$ throughout this interval, and its powers are monotone rather than oscillatory. Since $F(\rho) \sim 4\rho^5$, the remaining scalar propagation integral to infinity is finite. We stop at $\rho = 10$ within a regular chart; the figures do not test passage through a turning point or continuation through infinity. The complex initial modes do not impose $v_j = \overline{u_j}$.

The sigma formulas are evaluated at 33 equally spaced propagation values. Independent RK4 integrations of the full ten-component polynomial field use 32,768, 65,536 and 131,072 uniform steps over the same interval. Every comparison time is a node of each grid, so no interpolation is used. The finest integration supplies 513 saved states for the plotted lines. Table 3 reports the maximum of the scaled mode error over all ten components and all 33 observations, rather than only the endpoint.

Steps	Step size	Maximum mode error	Reduction factor
32,768	9.0341277×10^{-7}	$2.1807079 \times 10^{-20}$	—
65,536	4.5170639×10^{-7}	$1.3630311 \times 10^{-21}$	15.998960
131,072	2.2585319×10^{-7}	$8.5192239 \times 10^{-23}$	15.999475

Table 3: Extended genus-two trajectory at 30-digit working precision. Step halving gives the expected fourth-order reduction in RK4 error.

For the finest grid, the maximum scaled Hamiltonian and weighted-invariant drifts at the observation times are $7.6908187 \times 10^{-24}$ and $1.2009690 \times 10^{-22}$. The Hamiltonian drift is divided by $\max(1, |U|, |V|, |Q|)$ at each observation; the drift of $C_j = I_j - \beta_j I_5$ is divided by $\max(1, |\gamma_j - \beta_j \gamma_5|)$. These results distinguish arithmetic precision from finite-step integration accuracy.

The scalar coordinate at each observation is obtained by inversion of its signed quadrature, and the modes are then evaluated from their sigma expressions.

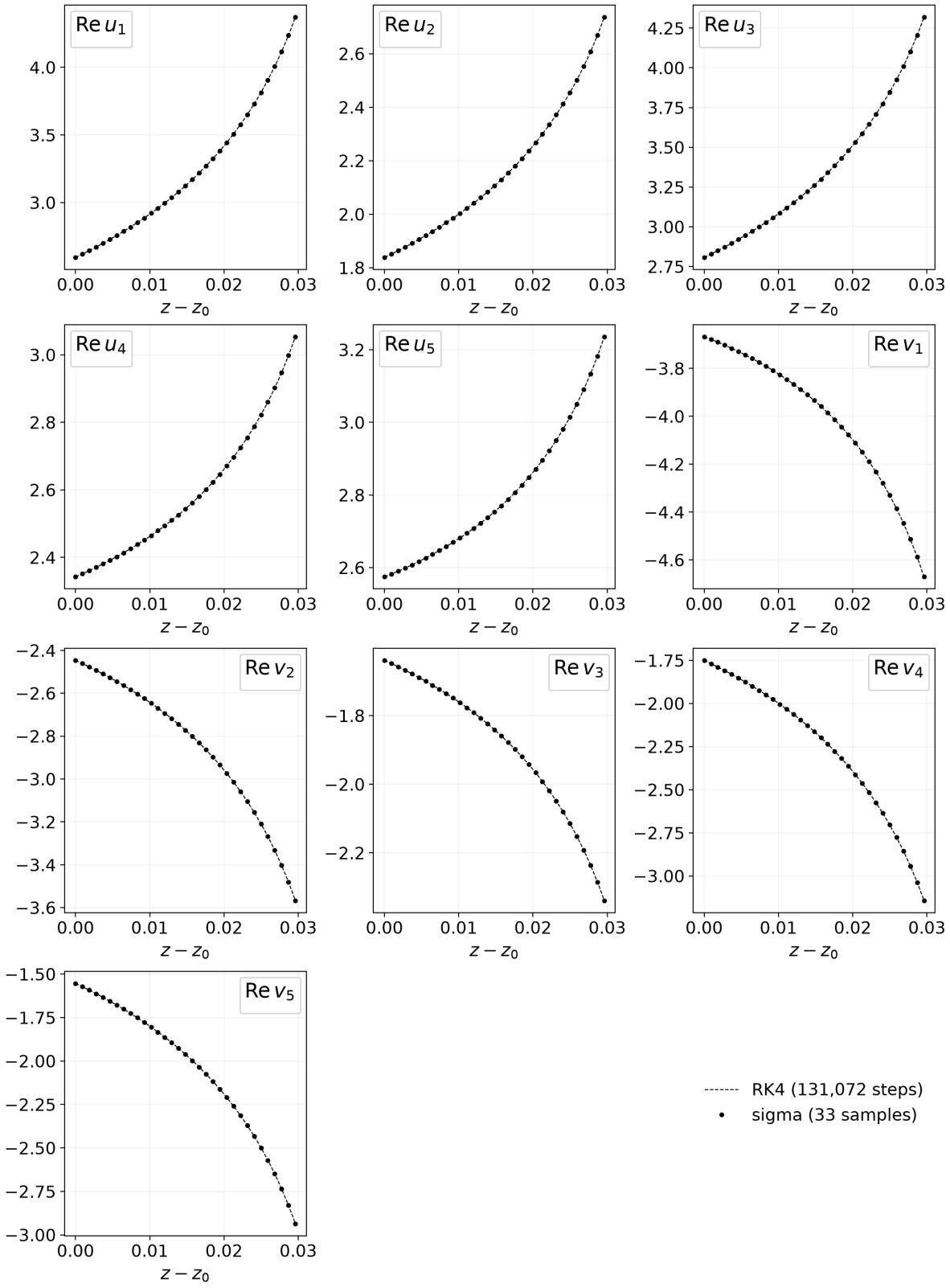


Figure 1: Real parts of all five genus-two u_j and v_j modes, ordered across rows with the u_j group first, on the positive-sheet trajectory $4.5 \leq \rho \leq 10$, $0 \leq z - z_0 \leq 0.02960303$. Thin dashed black lines show full-system RK4 with 131,072 steps; black markers show 33 evaluations of the respective sigma mode formulas. Actual amplitudes are shown without individual rescaling.

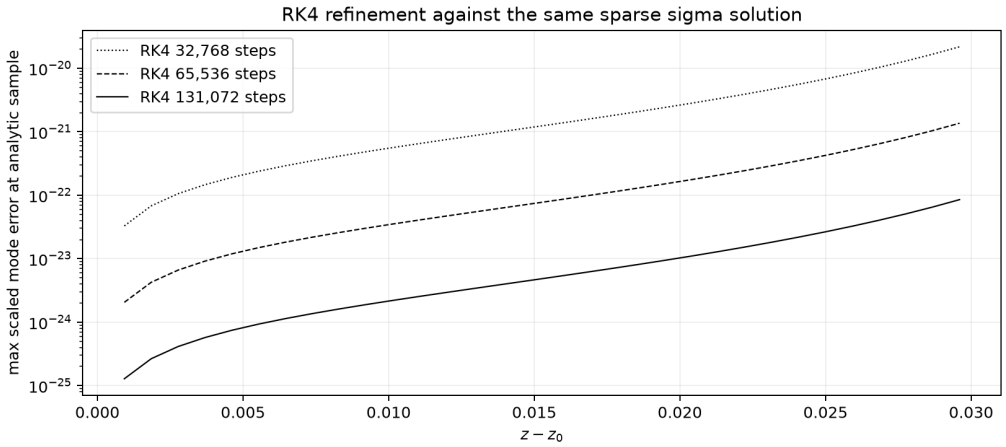


Figure 2: Maximum scaled discrepancy over the ten genus-two mode components at each noninitial observation time. The three RK4 grids are compared with the same 33 sigma evaluations; the initial discrepancy is omitted from the logarithmic plot.

7.2 Oscillatory genus-three trajectory

The genus-three example admits bounded scalar motion on the same Hamiltonian and invariant level. All seven roots of (76) are real and simple. Between the fifth and sixth roots,

$$L \approx 0.3505953510803235, \qquad R \approx 0.5739427200213239,$$

we have $F(\rho) \geq 0$. The scalar coordinate oscillates between these turning points, so the powers $I_j = \gamma_j - \beta_j \rho$ are periodic. Their period is

$$T = 2 \int_L^R \frac{d\rho}{\sqrt{F(\rho)}} \approx 6.624556082295094.$$

We follow three cycles, over $0 \leq z - z_0 \leq 3T \approx 19.87366824688528$.

We start at $\rho(z_0) = (L + R)/2$ with $\rho'(z_0) = +\sqrt{F(\rho(z_0))}$. The initial u_j are positive real numbers in the proportions

$$(u_1(z_0), \ldots, u_7(z_0)) \;\propto\; (1.1,\ 0.8,\ 1.2,\ 1,\ 1.1,\ 0.9,\ 1.05).$$

Their common scale is determined by the initial product condition

$$u_1(z_0)^2 \prod_{j=2}^{7} u_j(z_0) = \frac{q(\rho(z_0)) + \rho'(z_0)}{2}, \qquad v_j(z_0) = \frac{\gamma_j - \beta_j \rho(z_0)}{u_j(z_0)}.$$

The complete modes are then given by (77). The sixth and seventh powers cross zero at $\rho = 0.4$ and 0.55; the polynomial coupled equations remain regular there. These initial conditions do not impose $v_j = \overline{u_j}$, so the signed powers are not positive optical intensities.

For continuation, we follow the signed curve point $(\rho(z), \rho'(z))$ through each turning point and integrate the Abel differentials along that path, keeping $\Delta w_1 = z - z_0$. A regular curve parameter removes the square-root singularity of the ρ coordinate at a turning point. The Abel vector retains its accumulated periods as the point completes successive cycles. Both u_j and v_j are evaluated from their own sigma expressions, which allow mode zeros without division by a vanishing mode.

Figures 4–5 show the scalar coordinate, periodic powers and individual modes. The modes can have growing or decaying envelopes even when their products repeat: the sigma

expressions acquire reciprocal cycle multipliers for u_j and v_j. After three scalar cycles, for example, $u_6(z_0 + 3T)/u_6(z_0) \approx -1.0997562$ and $u_7(z_0 + 3T)/u_7(z_0) \approx -0.025748382$. The plots retain these envelopes and sign reversals.

Independent full-system RK4 integrations are compared with 97 equally spaced sigma evaluations on common grid nodes. Table 4 reports the maximum scaled mode discrepancy over all fourteen components and all 97 observations; Figure 3 shows its evolution along the trajectory.

Steps	Step size	Maximum mode error	Reduction factor
24,576	8.0866163×10^{-4}	$2.5450058 \times 10^{-13}$	—
49,152	4.0433082×10^{-4}	$1.5925384 \times 10^{-14}$	15.980813
98,304	2.0216541×10^{-4}	$9.9593318 \times 10^{-16}$	15.990414

Table 4: Full-system RK4 refinement over three genus-three scalar cycles. The same 97 sigma observations are used on all grids, at 30-digit working precision.

The step-halving factors are close to 16, as expected for fourth-order RK4. On the finest grid, the maximum scaled Hamiltonian and weighted-invariant drifts are $1.5017527 \times 10^{-17}$ and $3.6848896 \times 10^{-18}$, with the same scaling conventions as above and $C_j = I_j - \beta_j I_7$. The mode discrepancy of approximately 10^{-15} measures finite-step integration error over three cycles, separately from the precision of the sigma evaluations.

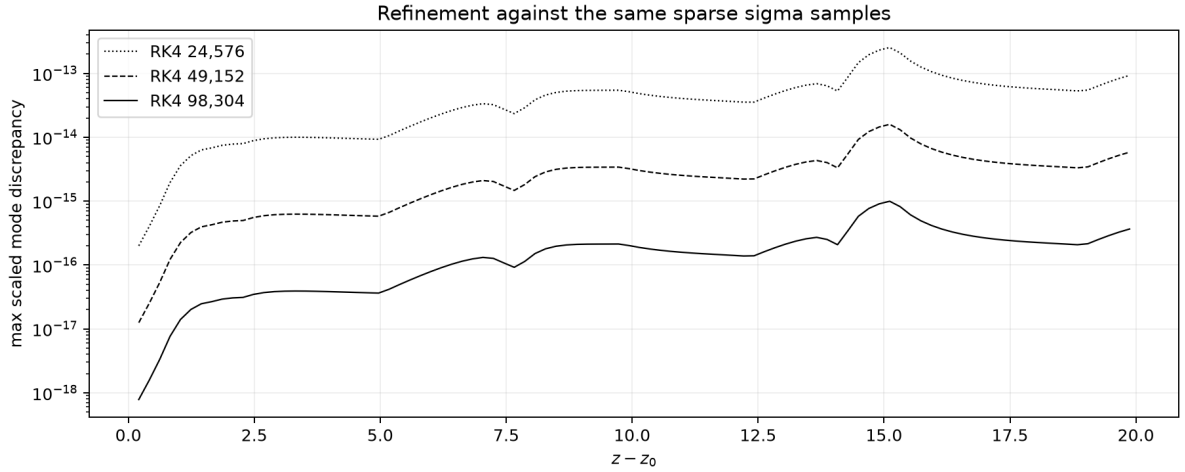


Figure 3: Scaled mode discrepancies over three genus-three scalar cycles for the three uniform RK4 grids. Each curve is the maximum over fourteen components at the same noninitial sigma observation times. Step halving reduces the maximum discrepancy by approximately 16.

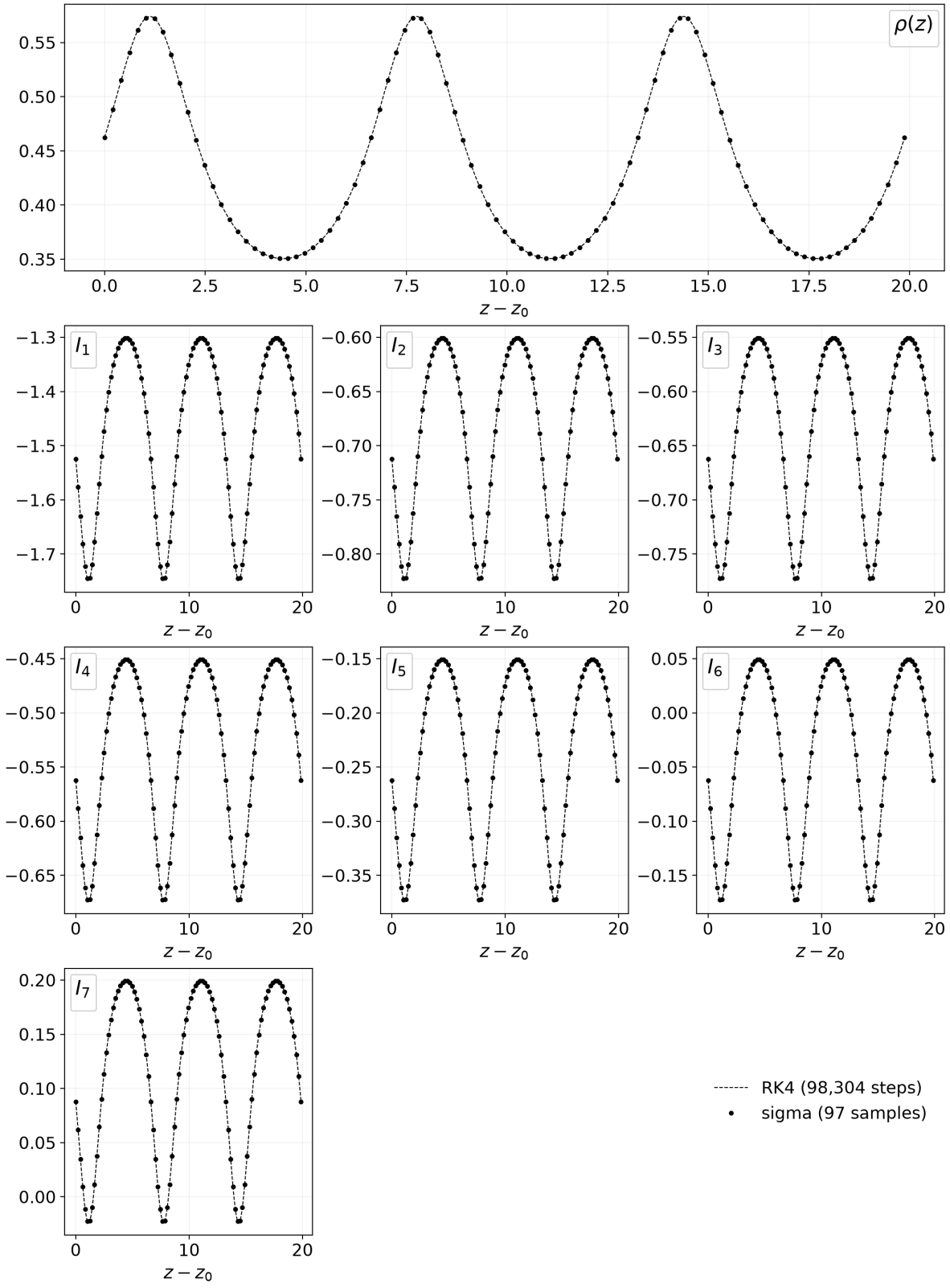


Figure 4: Genus-three scalar coordinate ρ (top) and all seven signed powers I_j over three cycles, $0 \le z - z_0 \le 19.87366825$. Their common oscillations show the affine relations $I_j = \gamma_j - \beta_j \rho$; the sixth and seventh powers cross zero. Thin dashed black lines show full-system RK4 with 98,304 steps; black markers show 97 sigma observations, using the coordinate ratio for ρ and the mode products for I_j.

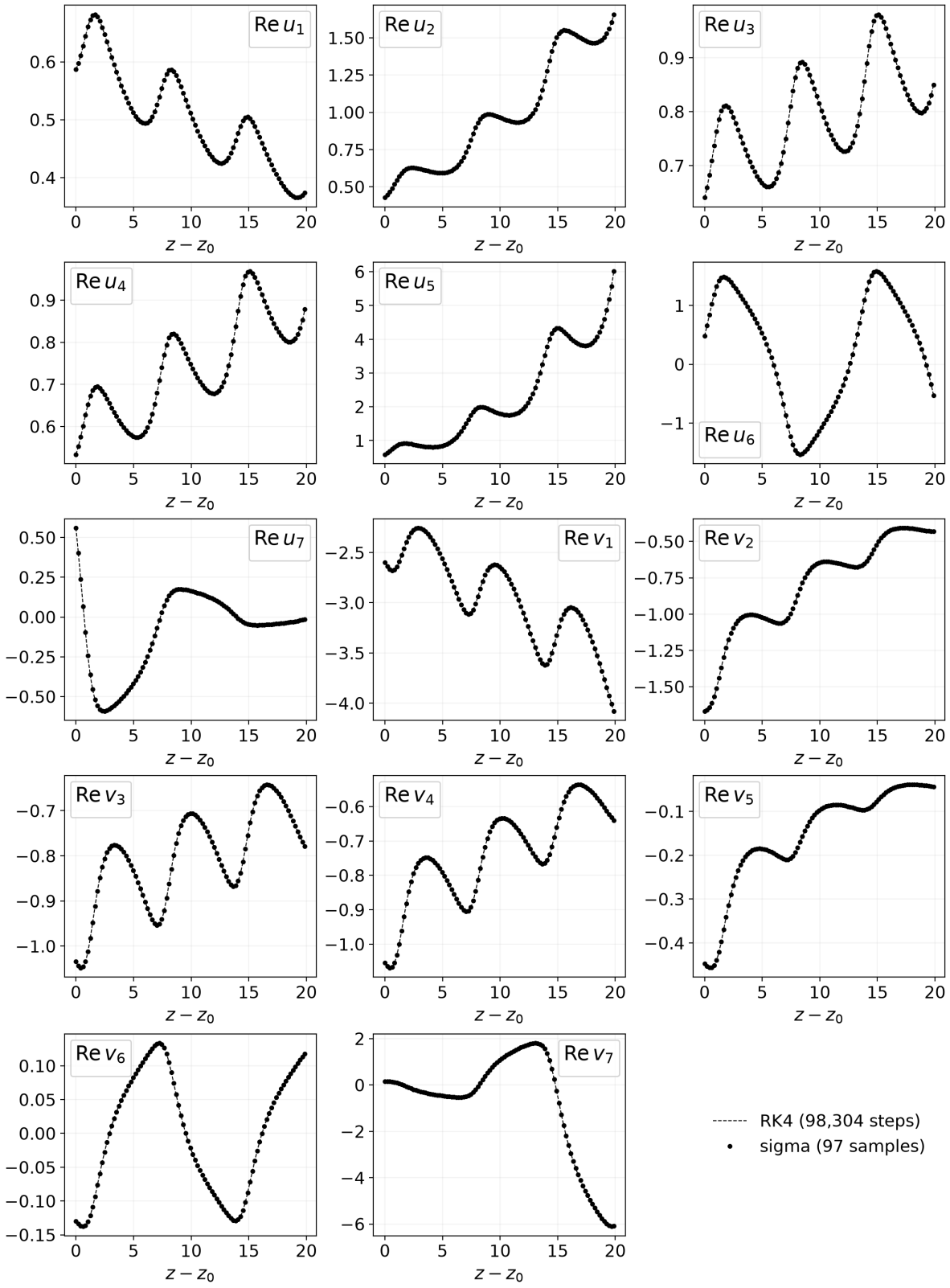


Figure 5: Real parts of all seven genus-three u_j and v_j modes over the same three scalar cycles as Figure 4, ordered across rows with the u_j group first. Thin dashed black lines show full-system RK4; black markers show direct sigma mode evaluations. Actual amplitudes retain their cycle multipliers, envelopes and sign reversals.

8 Discussion

The explicit solutions raise two further questions: what the conserved quantities establish about integrability, and how the local formulas behave beyond a chosen chart. We separate the general analytic construction from the scope of the examples and identify the remaining continuation and spectral problems.

8.1 Scope of the construction

The construction combines scalar reduction with individual-mode integration for the polynomial Hamiltonian class considered here. It requires a smooth odd-degree curve of leading coefficient four, distinct nonbranch fixed mode points, a regular mode chart, and $\deg B_j \le g-1$. A polynomial Q of degree at most g automatically satisfies the last condition; higher-degree polynomials may do so by cancellation, as both examples show. Integer weights, including a double-weight mode, enter through the power line and weighted products.

Regularized second-kind integrals define the exponential coefficients in arbitrary genus. The short fixed-point sigma expressions and full canonical trajectory checks in this paper cover genera two and three. Scalar coordinate recovery is given by the arbitrary-genus derivative ratio, with an additional genus-four formula and a separate check at regular curve points. Finite sigma-based coefficient extraction is available in every genus; uniform short fixed-point expressions remain a separate question. Singular curves and even-degree models require further treatment.

For genus one the same structural split gives a shifted Weierstrass sigma ratio with a zeta exponential. Its sole Abel coordinate still satisfies $w_1(z) = w_1(z_0) + z - z_0$. In higher genus only w_1 evolves linearly, the one-point image lies on the theta divisor, and sharp/flat derivatives provide the mode kernel. The scalar inversion ratios are separate restrictions to that image.

8.2 Commuting integrals and local integrability

The weighted powers in (3), together with H, give N commuting integrals. With the canonical bracket

$$\{A, B\} = \sum_{j=1}^{N} \left(\partial_{u_j} A \, \partial_{v_j} B - \partial_{v_j} A \, \partial_{u_j} B \right),$$

the powers commute mutually and satisfy $\{I_j, H\} = \beta_j(V-U)$. Hence $\{C_j, H\} = \{C_j, C_k\} = 0$.

On the open set where every $I_j \neq 0$, the differentials dI_j are independent and the $N-1$ weighted combinations dC_j have rank $N-1$. The tangent variation $u_1 \mapsto e^t u_1$, $v_1 \mapsto e^{-t} v_1$ preserves every power but gives $dH/dt = \beta_1(U-V)$. Therefore dH is independent of the dC_j where $U \neq V$. This proves complex local Liouville integrability on that open set; algebraic complete integrability is not established here. The short regular trajectory charts satisfy this condition. The oscillatory genus-three path also reaches turning points and modal zeros outside that open set; the independence argument above applies to its regular segments. A real compact-torus interpretation would additionally require reality conditions, regularity and compactness of the real invariant levels.

8.3 Continuation and spectral questions

The complete mode expressions resemble Baker–Akhiezer formulas, but here they are local functions on chosen Abel lifts. Exponentials of Abel integrals and sigma quasi-periods can produce multipliers under continuation; global meromorphicity on the compact curve is not asserted. Determining their monodromy, period structure, and a corresponding spectral

problem or Lax representation remains open. The scalar system is regular at finite turning points. The three-cycle calculation in Section 7.2 illustrates the resulting mode multipliers on one real oval; their structure on all cycles of the curve remains to be determined.

For repeated weights, the number of coordinate pairs N differs from the total weight $\sum_{j=1}^{N} \beta_j = \deg r$; the examples exploit that distinction and polynomial cancellation. It would be useful to characterize the most economical mode counts compatible with the smoothness, fixed-point, and correction-degree assumptions required here.

Further questions concern short sigma-only formulas for the higher-genus second-kind primitives, determinant descriptions beyond the interpolation families of Section 5, and conditions on Q, γ_j, β_j compatible with real positive powers and physical conjugacy. The explicit low-genus examples provide a starting point for those questions while retaining a precise local domain for the proved formulas.

Code availability

The project repository, https://github.com/HeskethGD/hyp-ham-sys, contains the computational notebooks, numerical verification scripts and figure-generation routines used in this work, together with the manuscript source, executed notebook exports and saved numerical results. The software is distributed under the MIT license, with its scope specified in the repository; that license does not apply to the manuscript or its figures. Dependency versions and execution instructions are recorded in the repository.

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